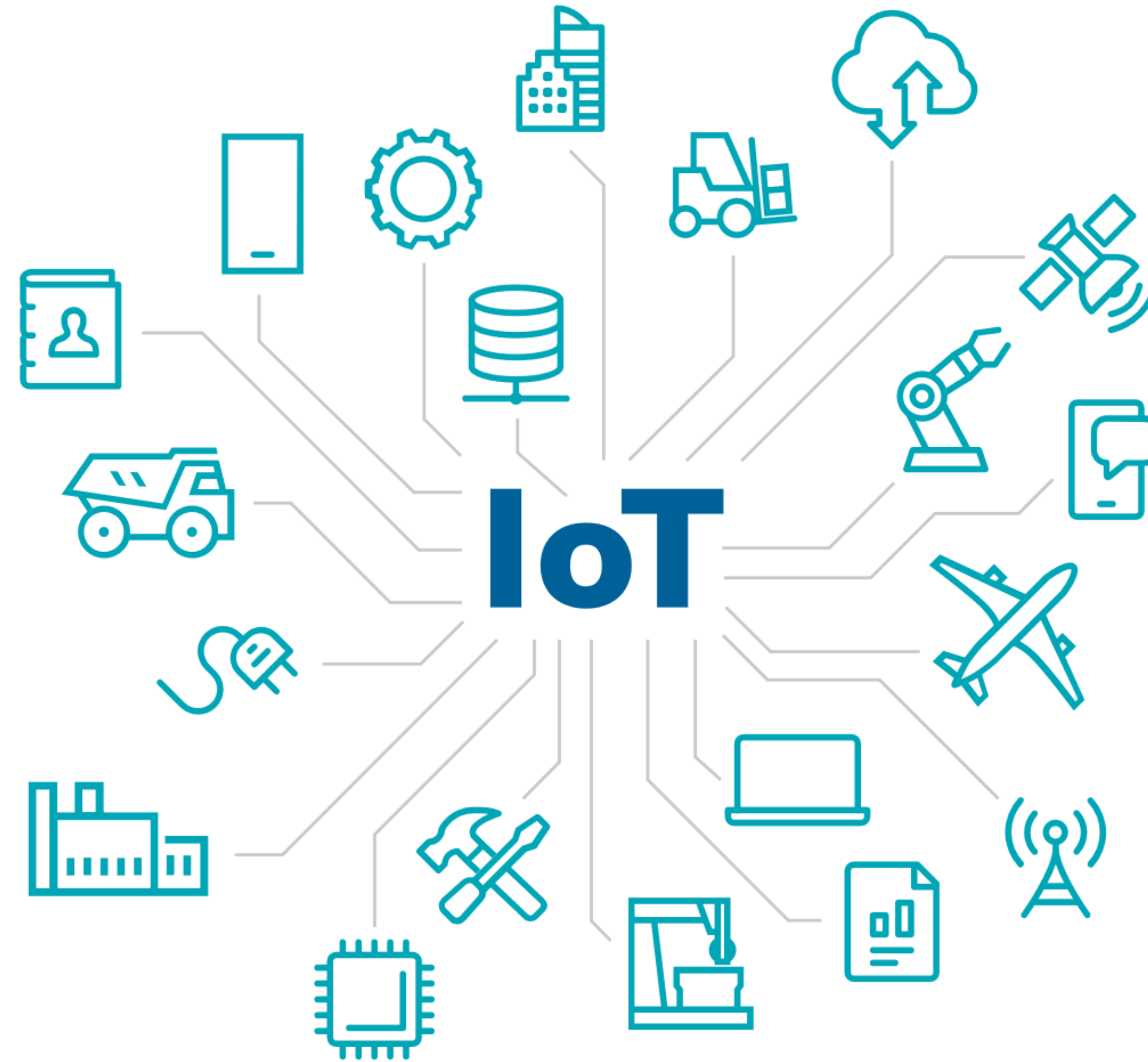
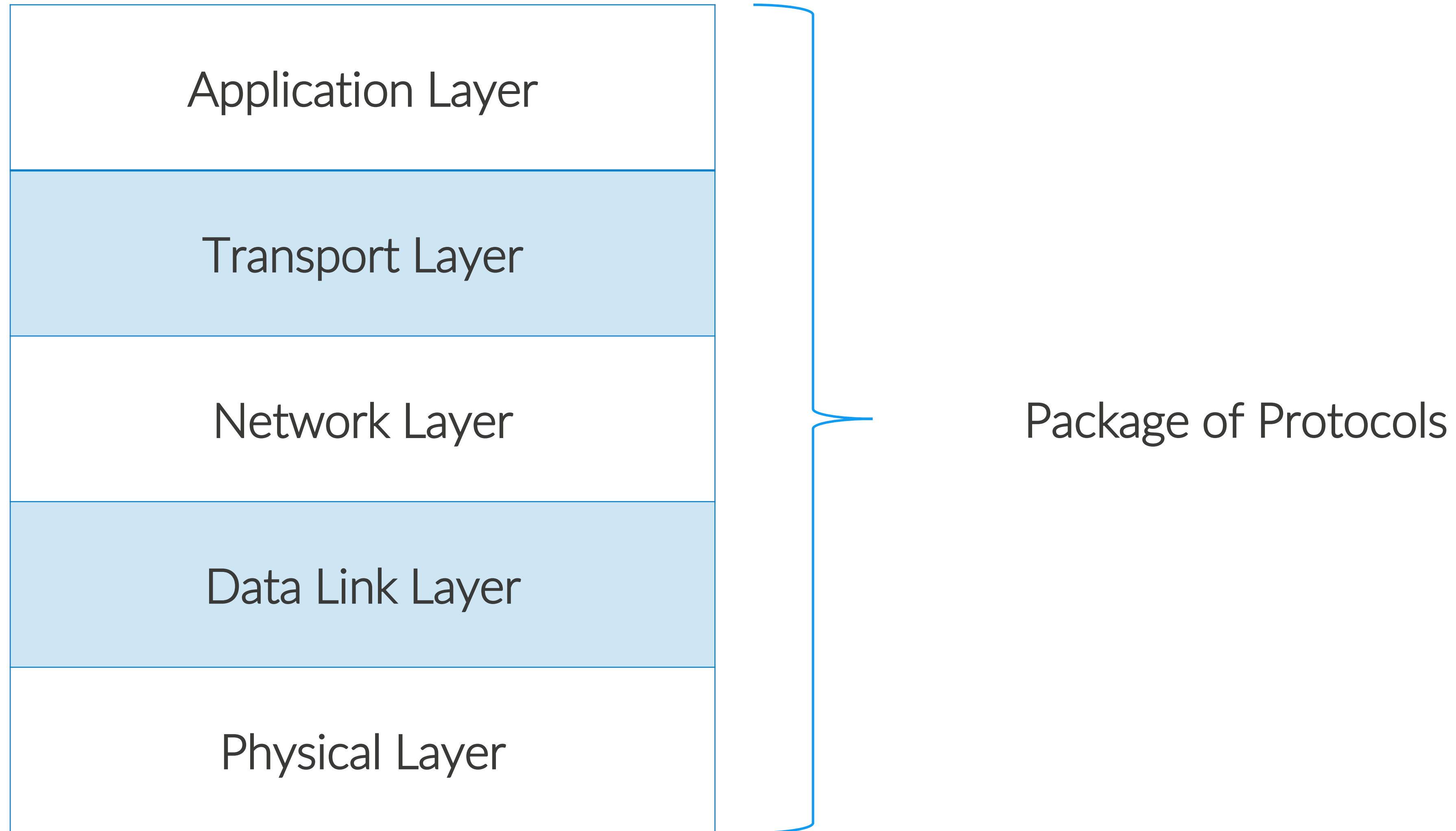




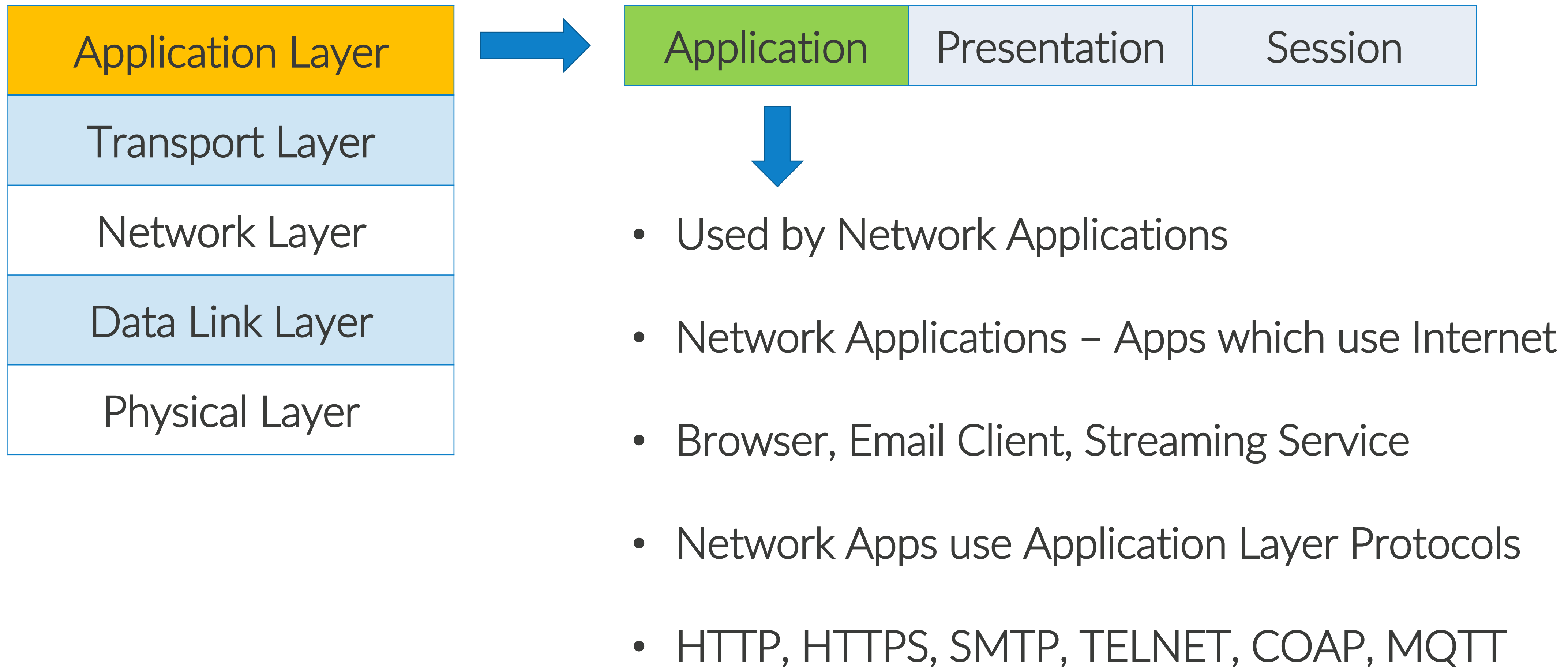
Module 2 – IoT Networks and Communications

- Dr. Abhishek Joshi

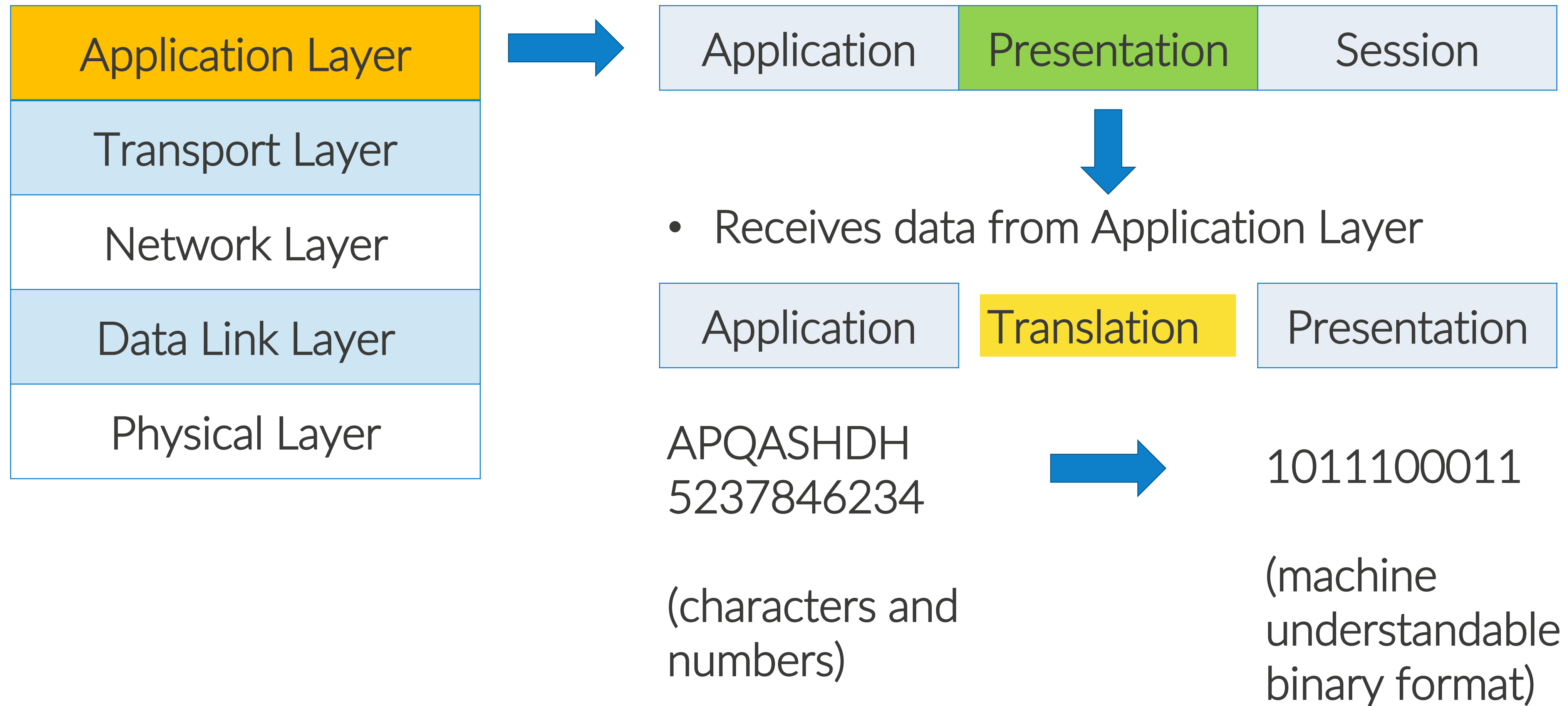
Protocol Stack for IoT Network



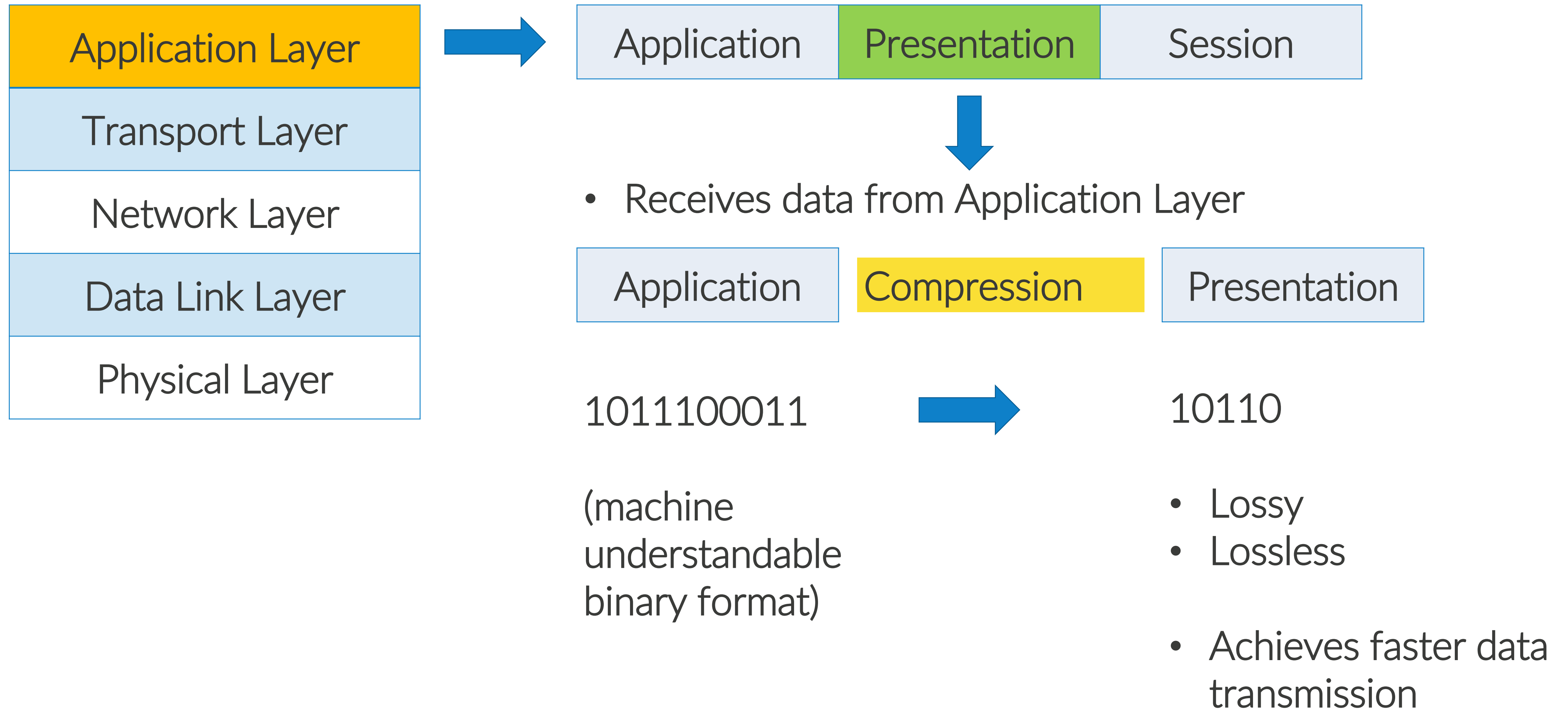
Protocol Stack : Application Layer



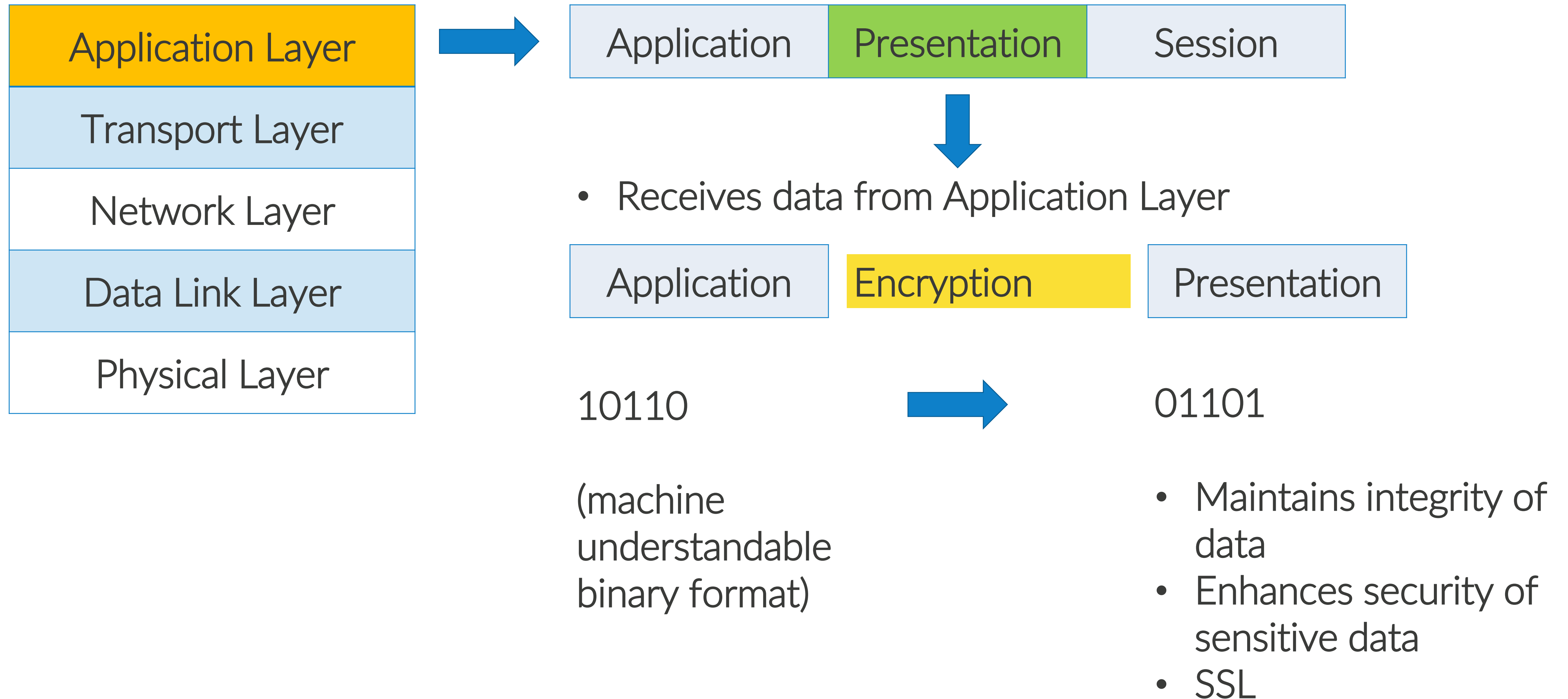
Protocol Stack : Application Layer



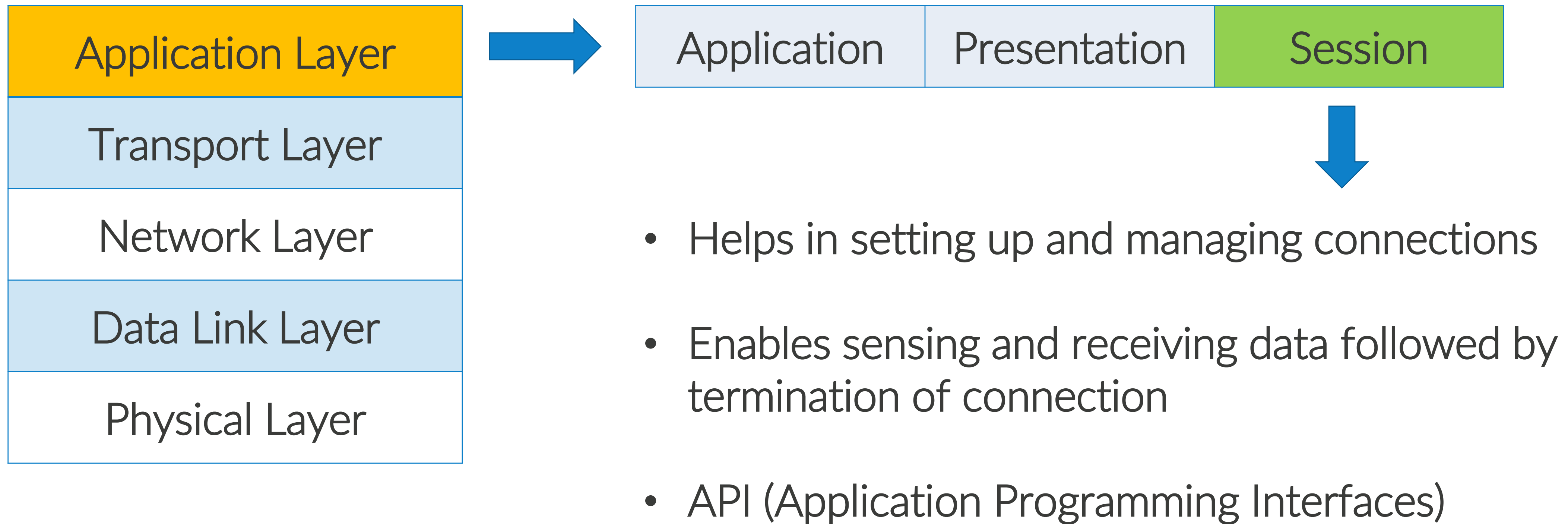
Protocol Stack : Application Layer



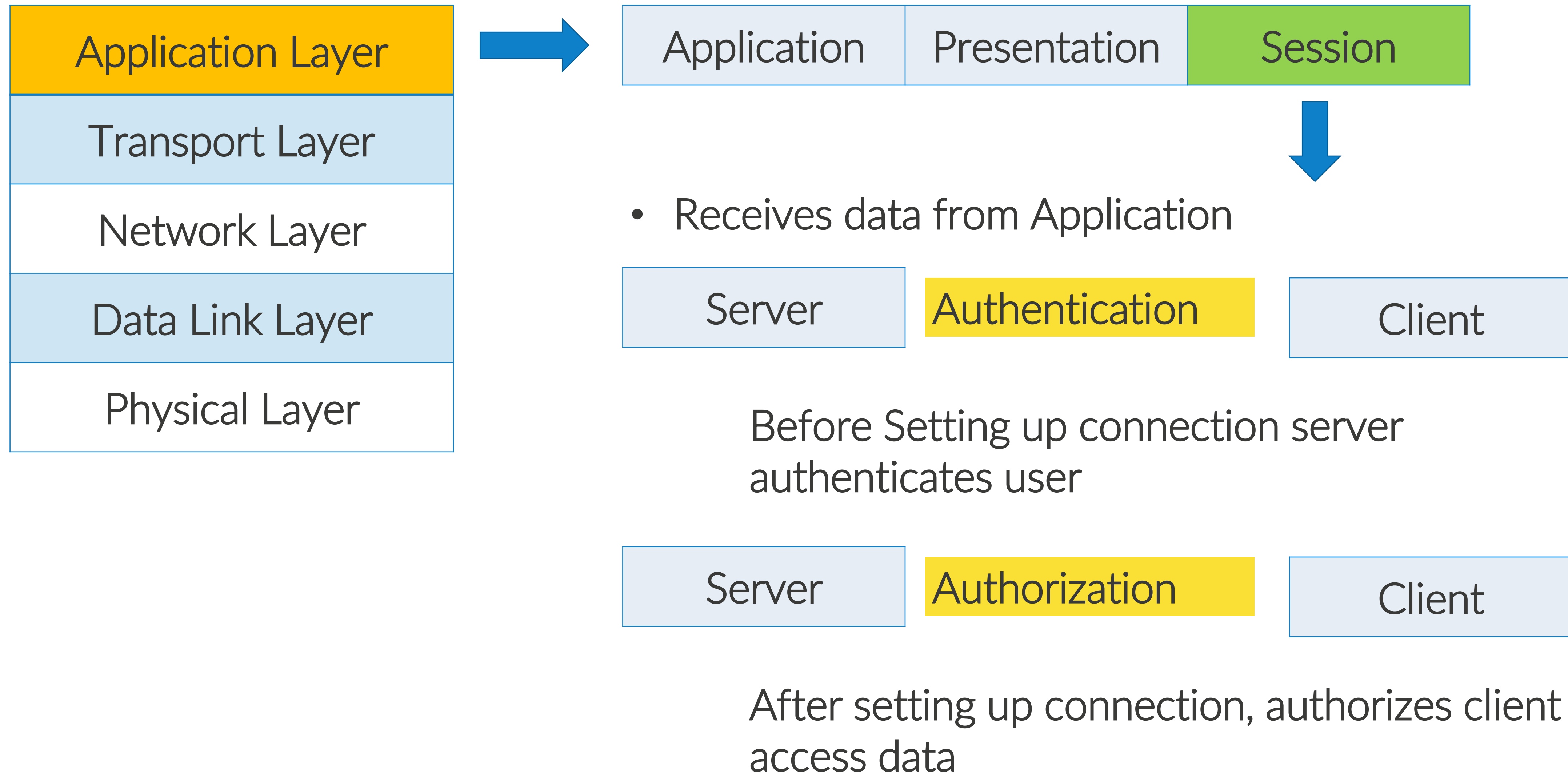
Protocol Stack : Application Layer



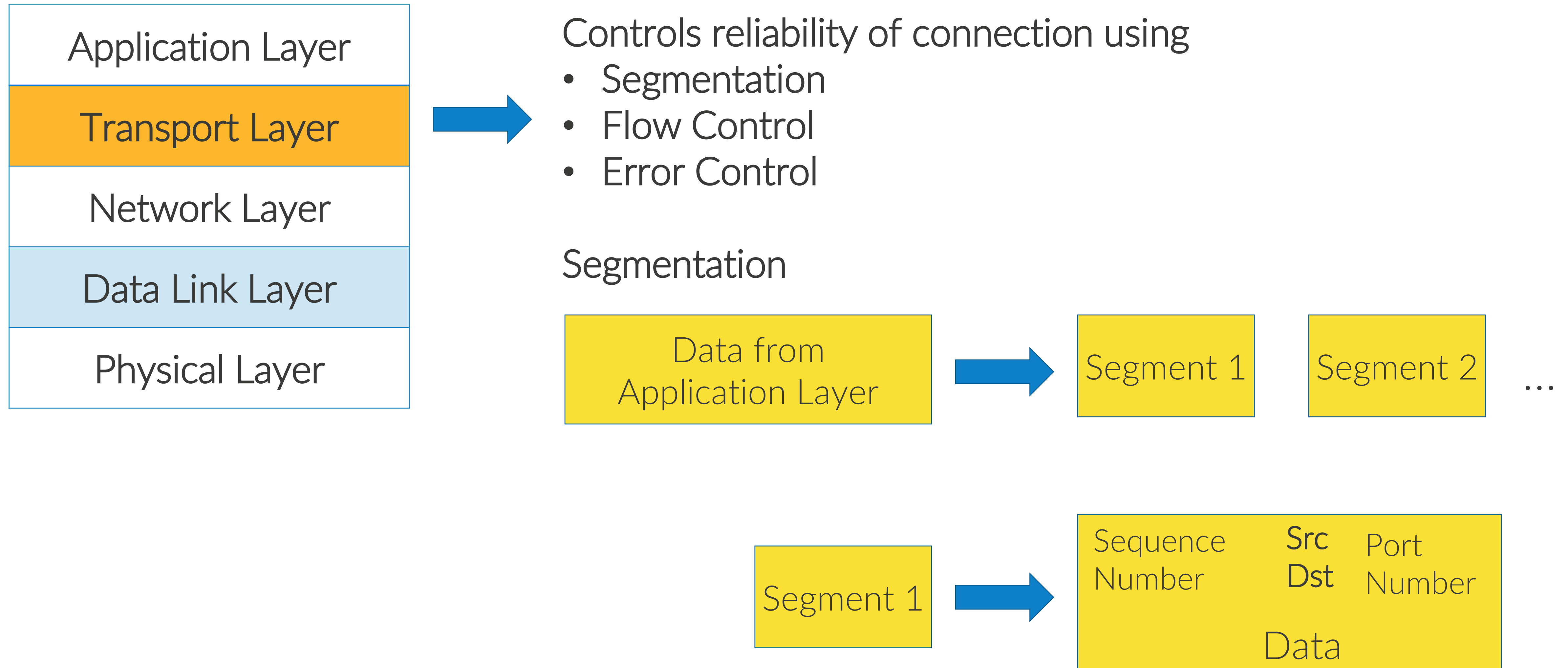
Protocol Stack : Application Layer



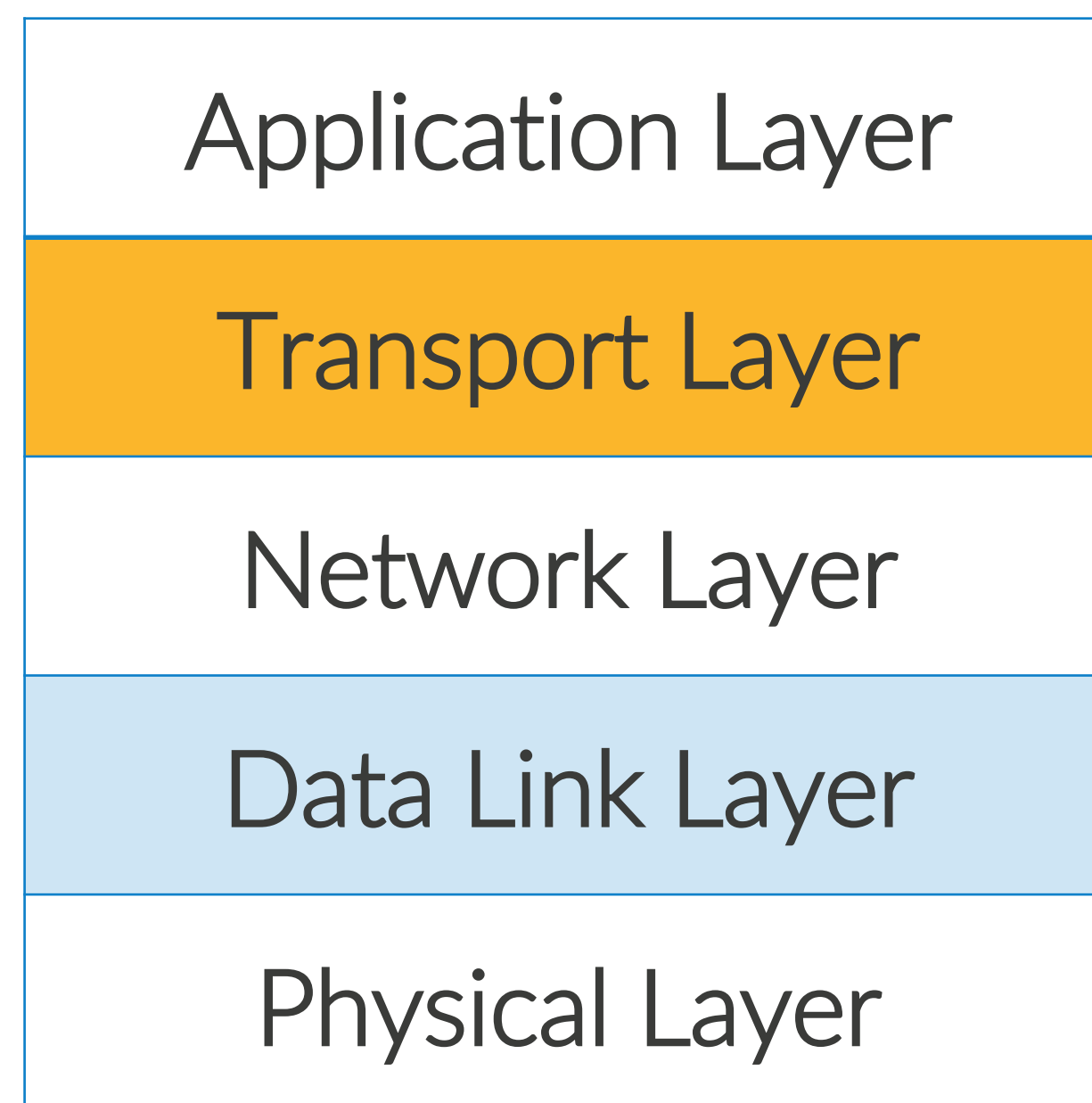
Protocol Stack : Application Layer



Protocol Stack : Application Layer



Protocol Stack : Transport Layer



Controls reliability of connection using

- Segmentation
- Flow Control
- Error Control

Flow Control

Controls amount of data to be transmitted

Server

100 Mbps

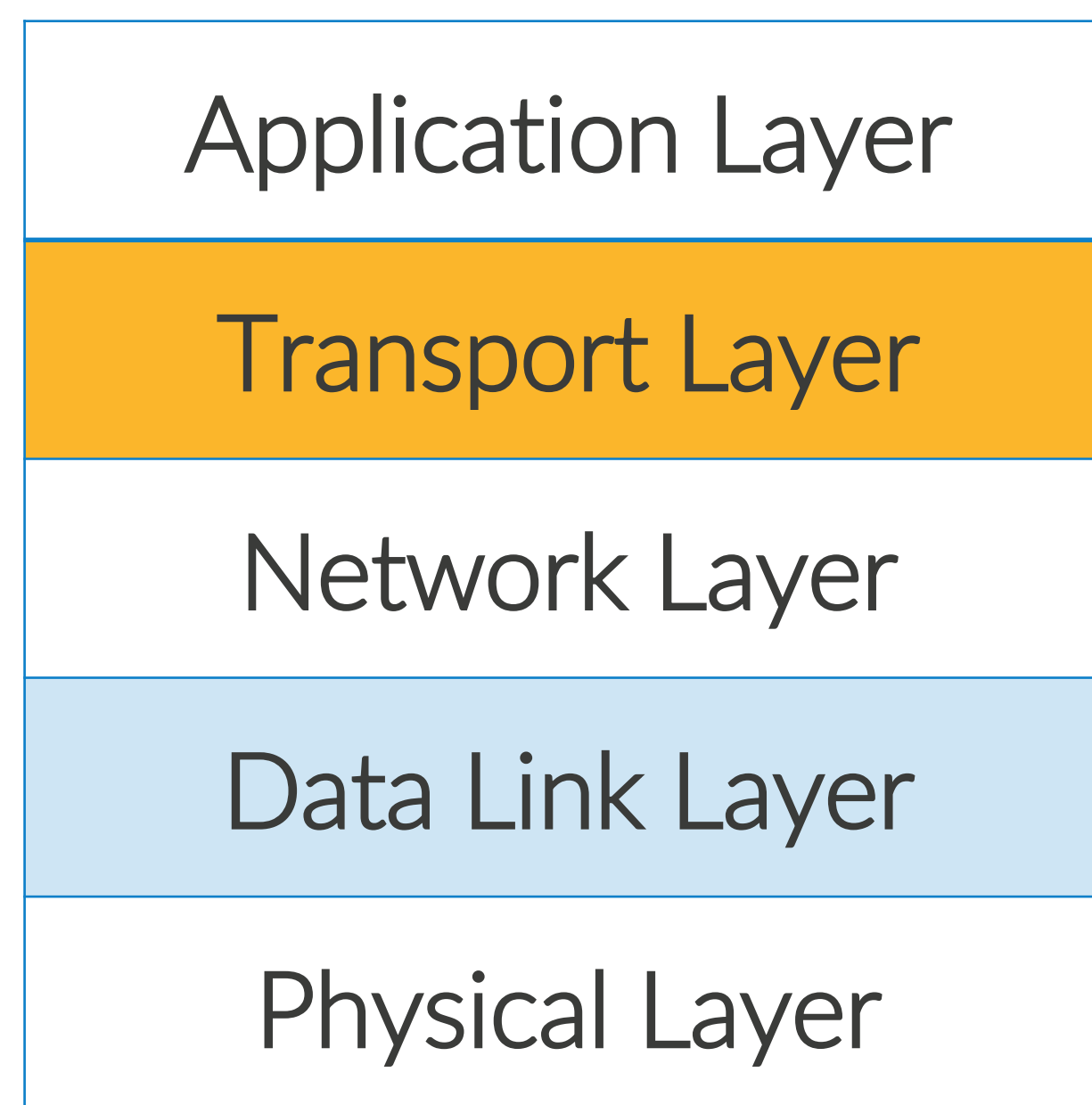
(Can Transmit Data)

Client

10 Mbps

(Can Process Data)

Protocol Stack : Transport Layer

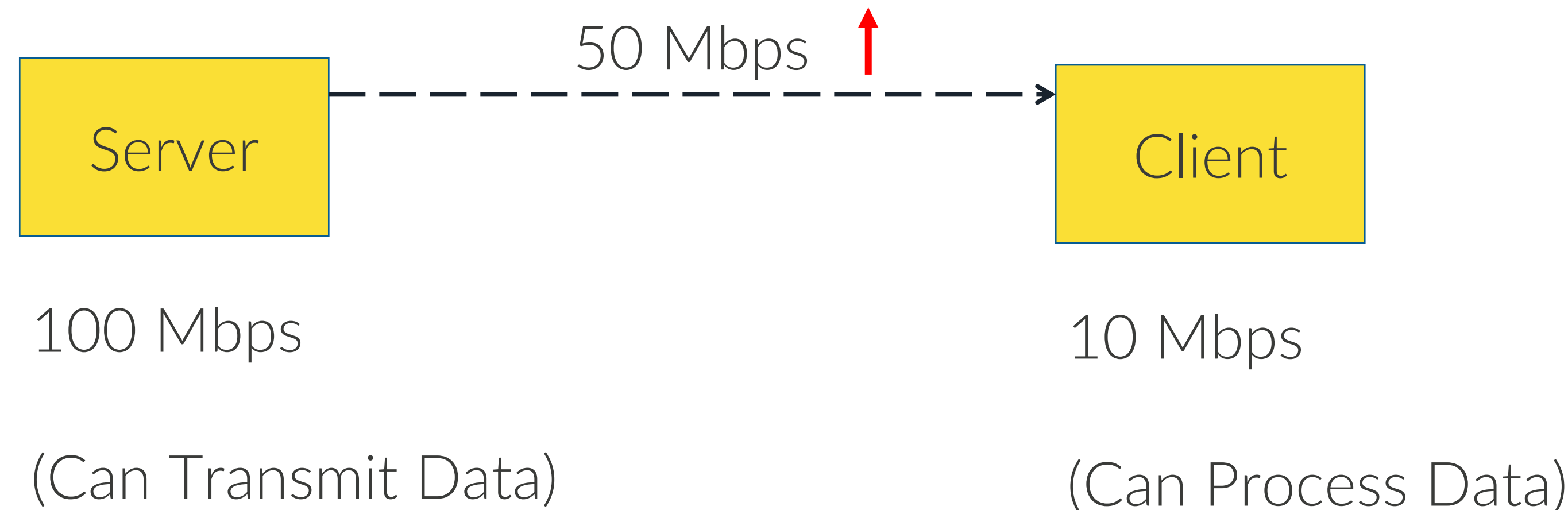


Controls reliability of connection using

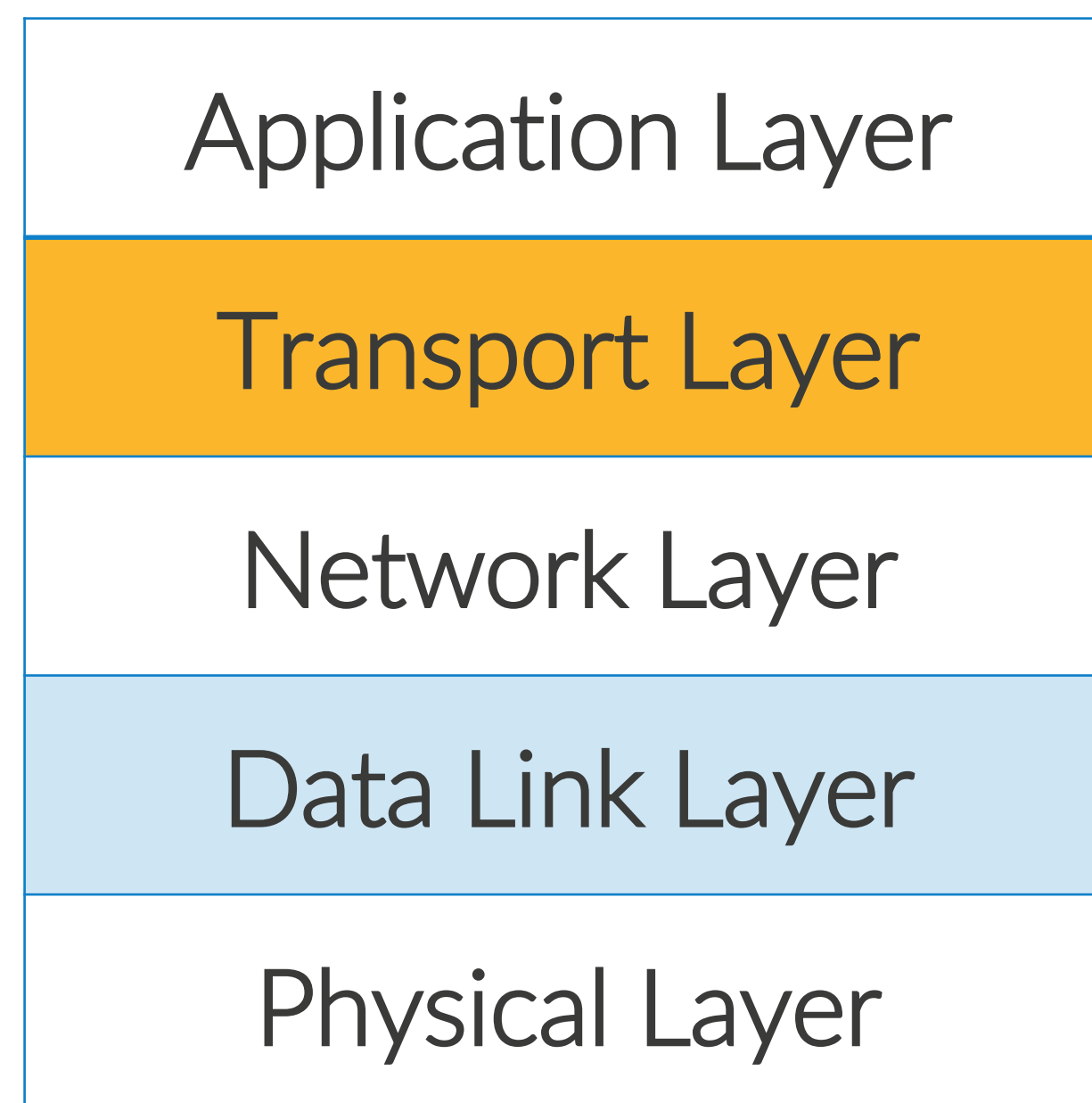
- Segmentation
- Flow Control
- Error Control

Flow Control

Controls amount of data to be transmitted



Protocol Stack : Transport Layer

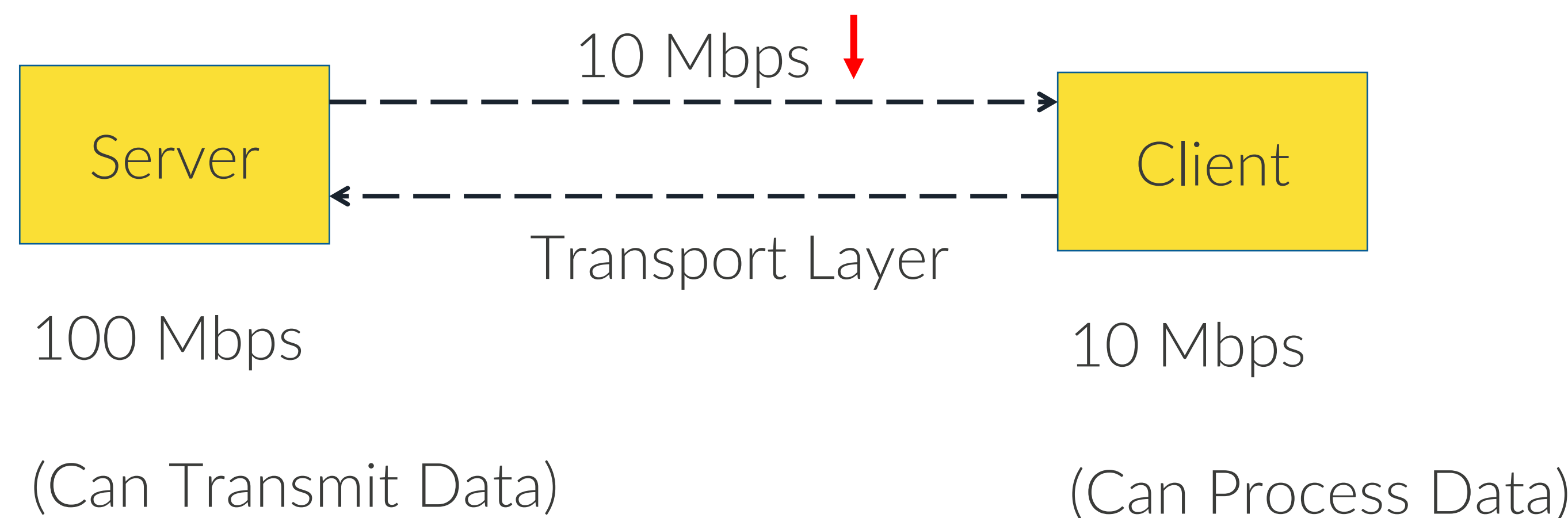


Controls reliability of connection using

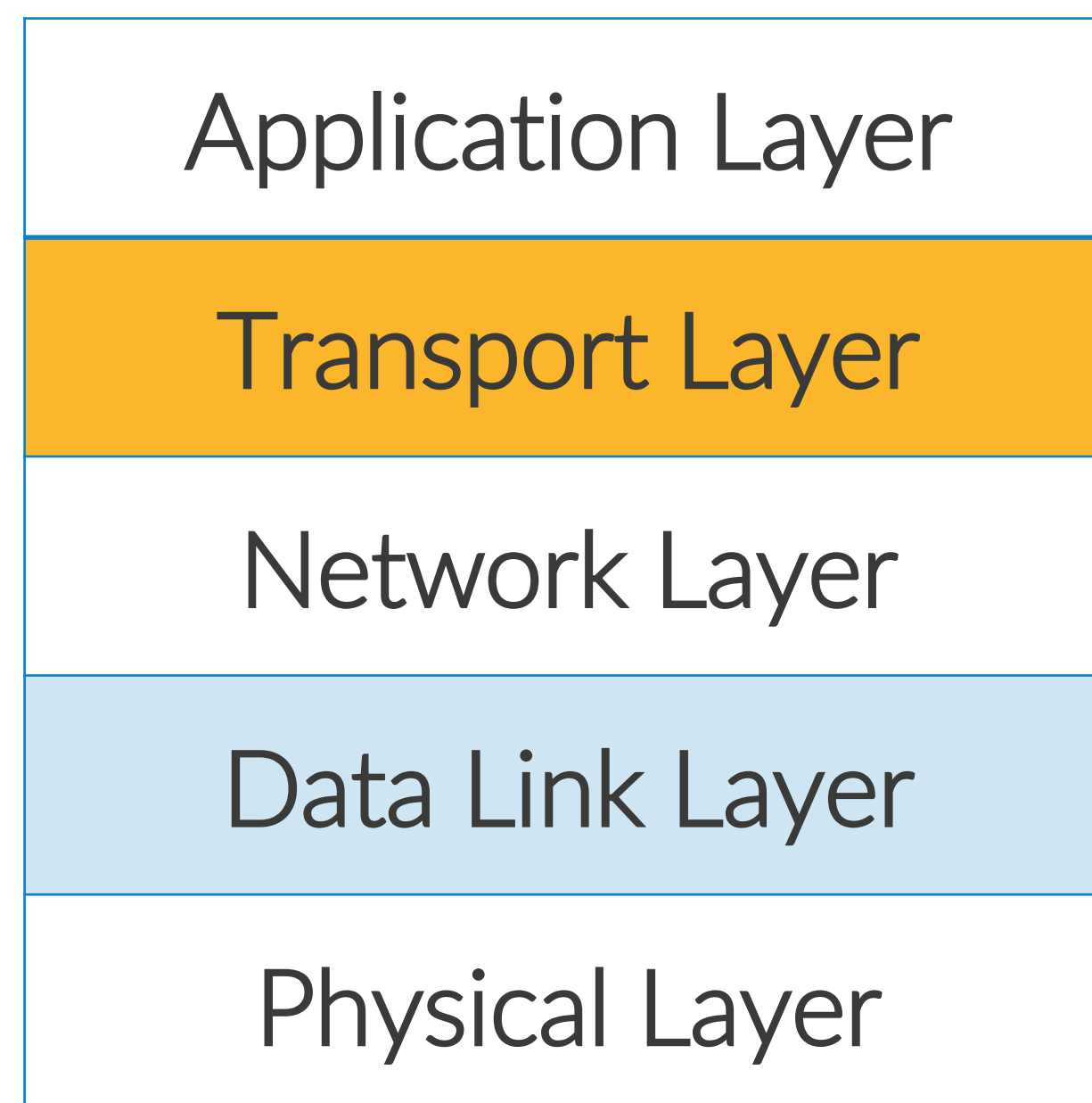
- Segmentation
- Flow Control
- Error Control

Flow Control

Controls amount of data to be transmitted



Protocol Stack : Transport Layer

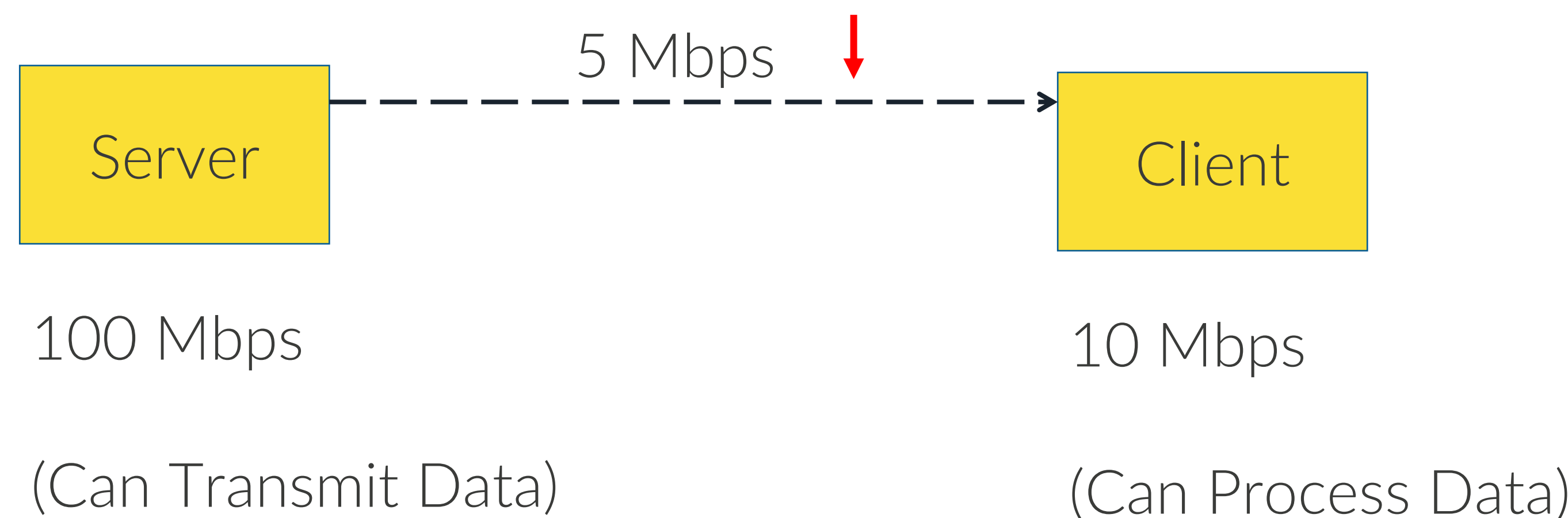


Controls reliability of connection using

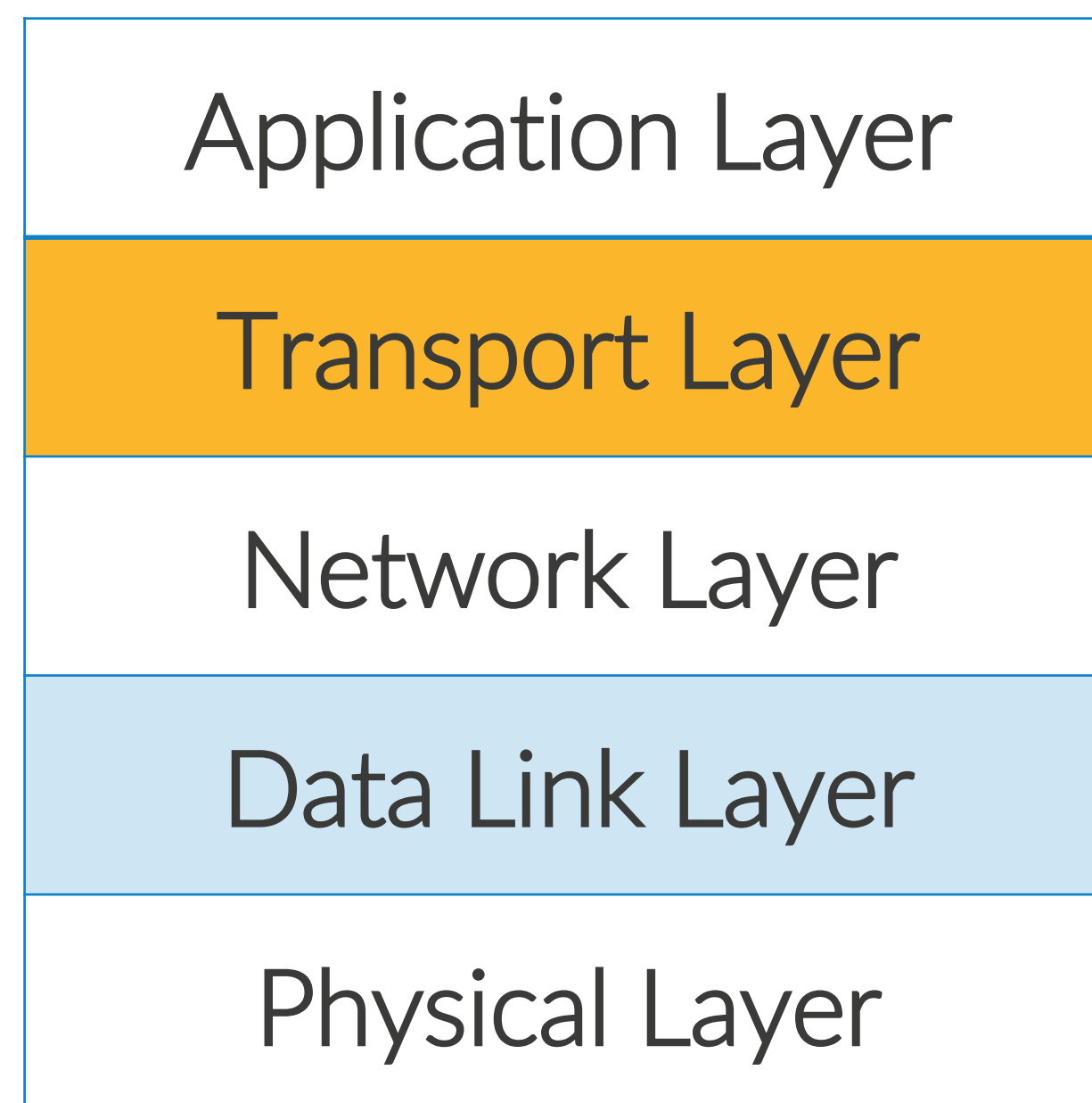
- Segmentation
- Flow Control
- Error Control

Flow Control

Controls amount of data to be transmitted



Protocol Stack : Transport Layer

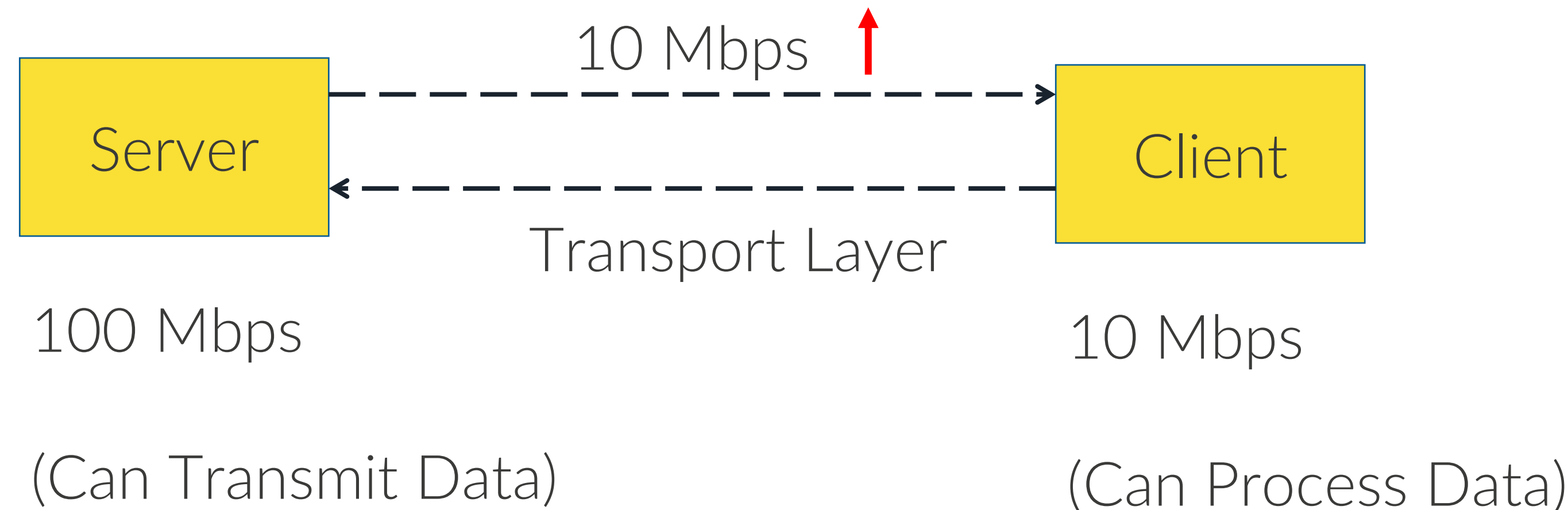


Controls reliability of connection using

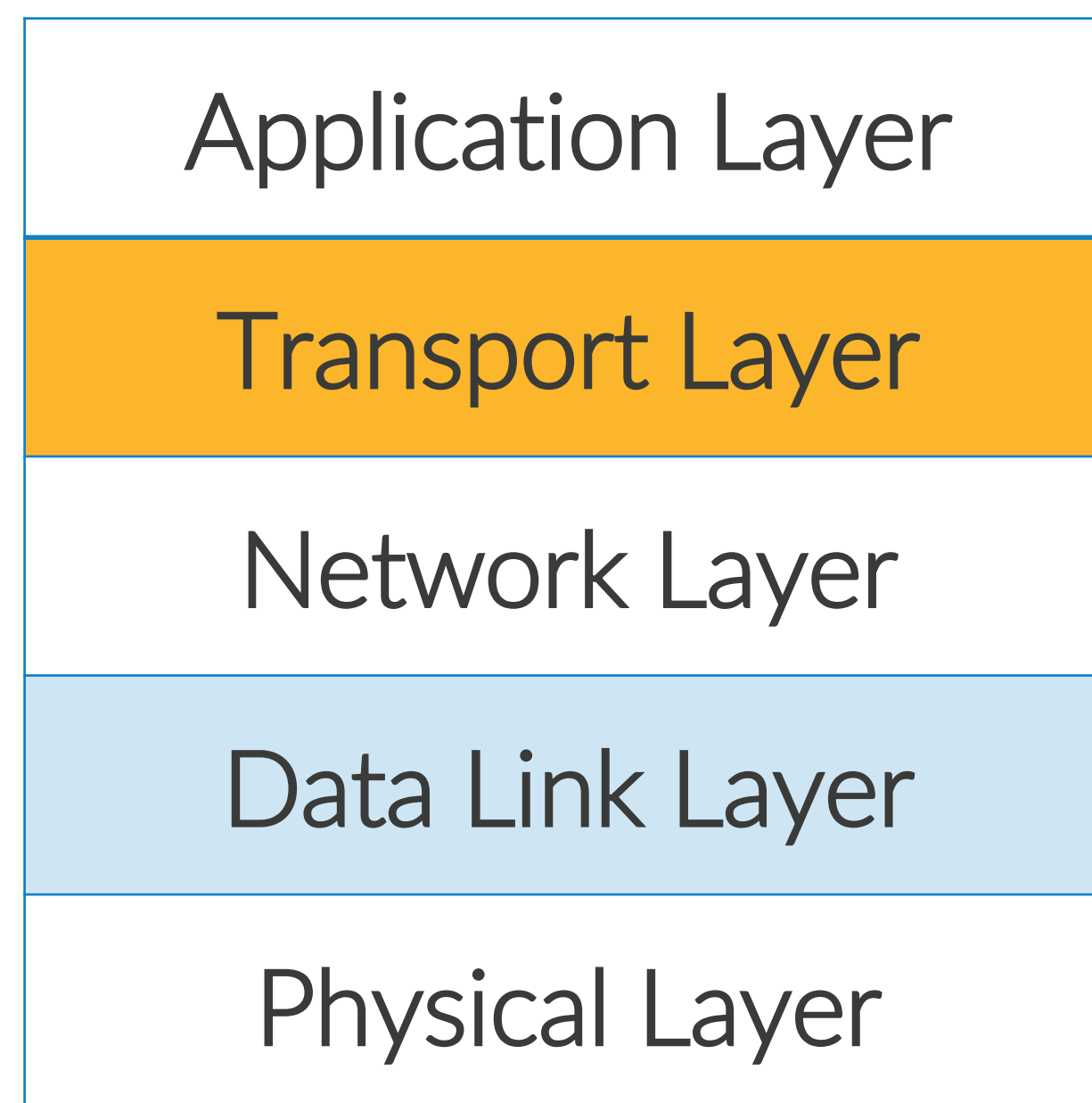
- Segmentation
- Flow Control
- Error Control

Flow Control

Controls amount of data to be transmitted



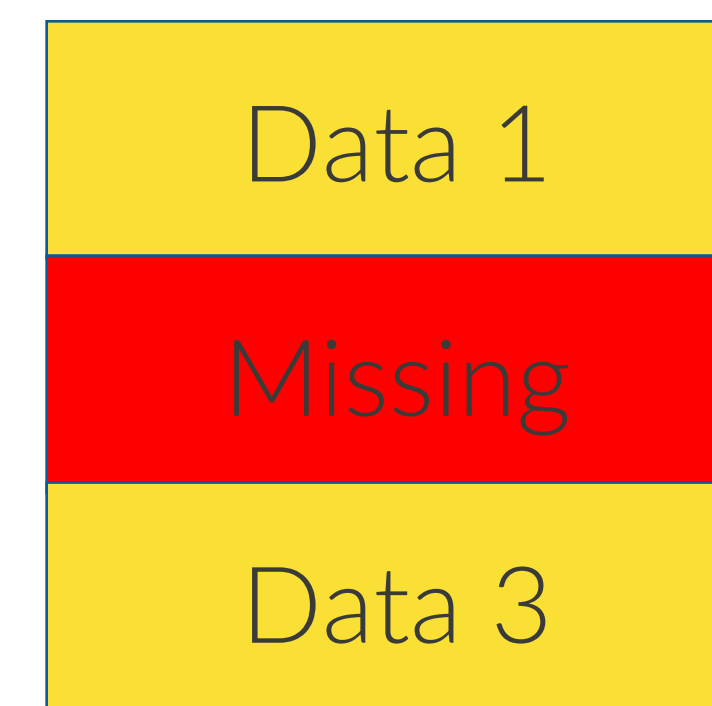
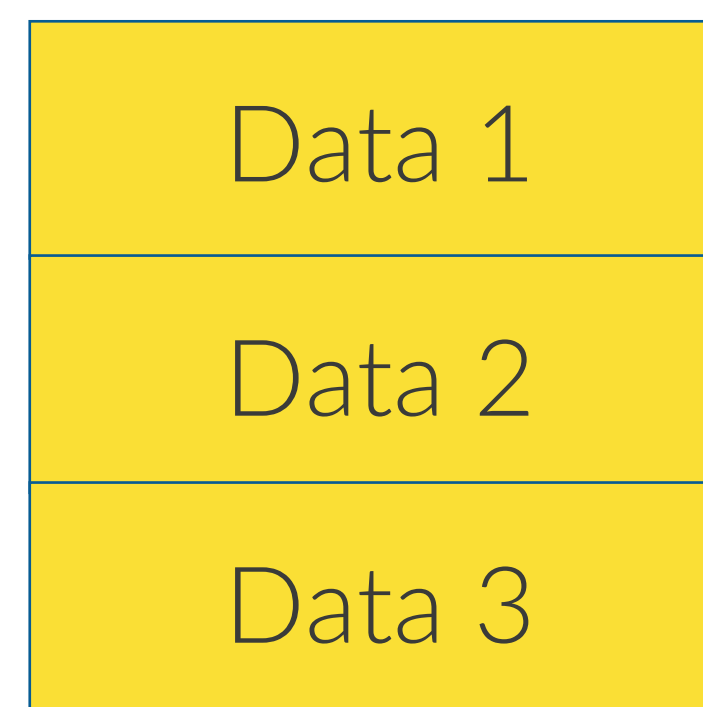
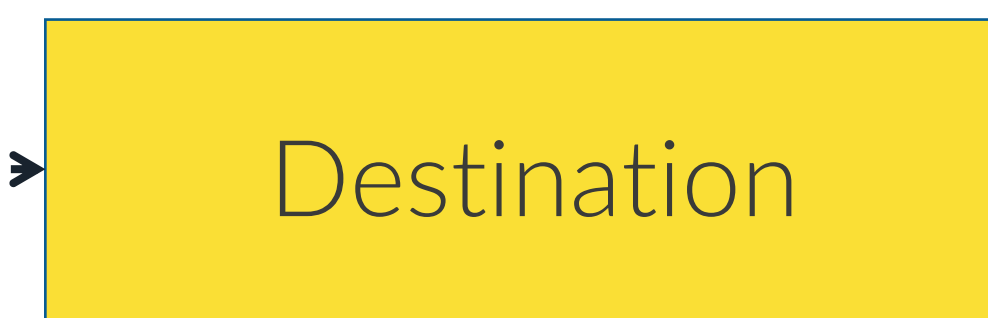
Protocol Stack : Transport Layer



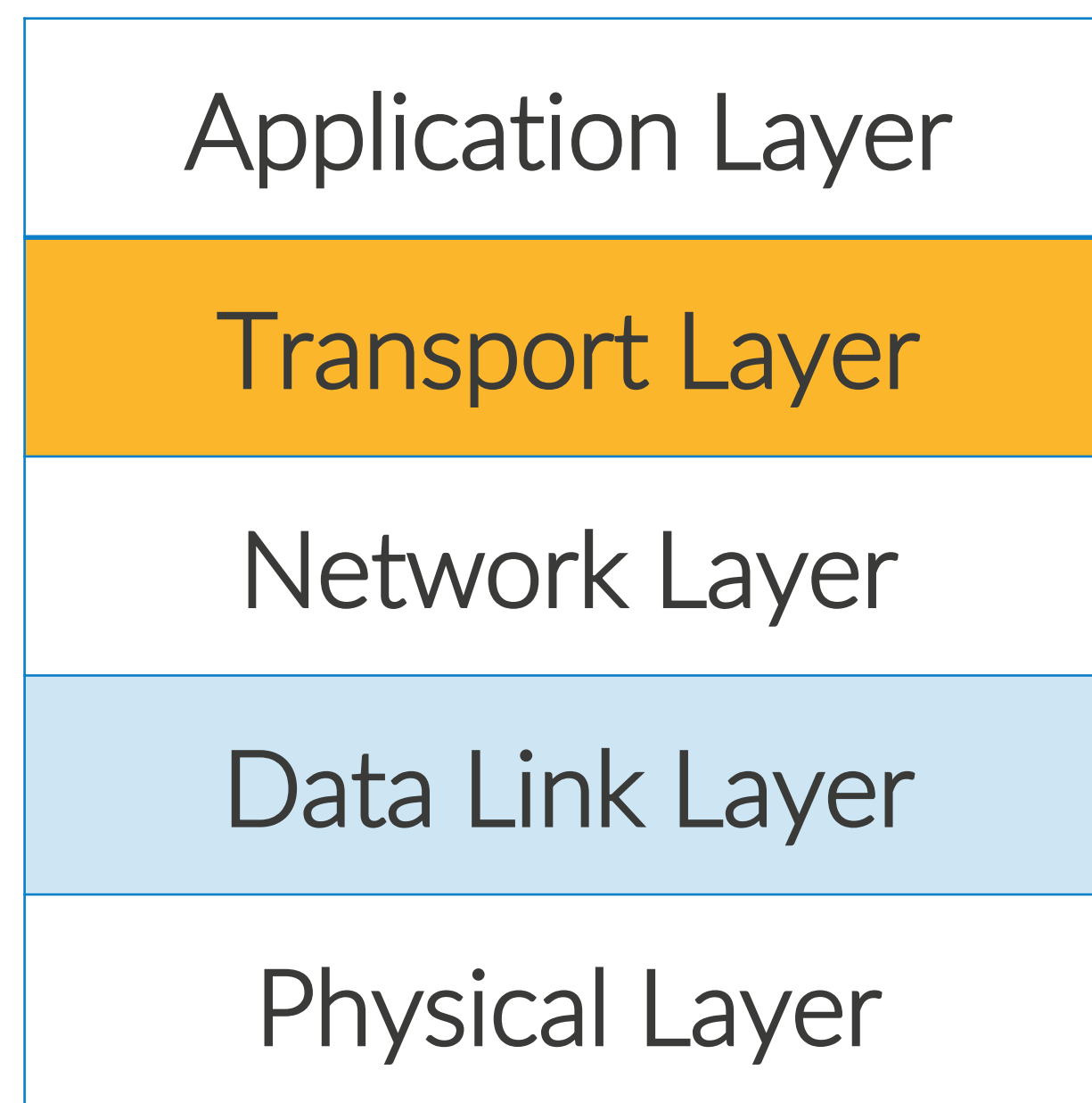
Controls reliability of connection using

- Segmentation
- Flow Control
- Error Control

Error Control



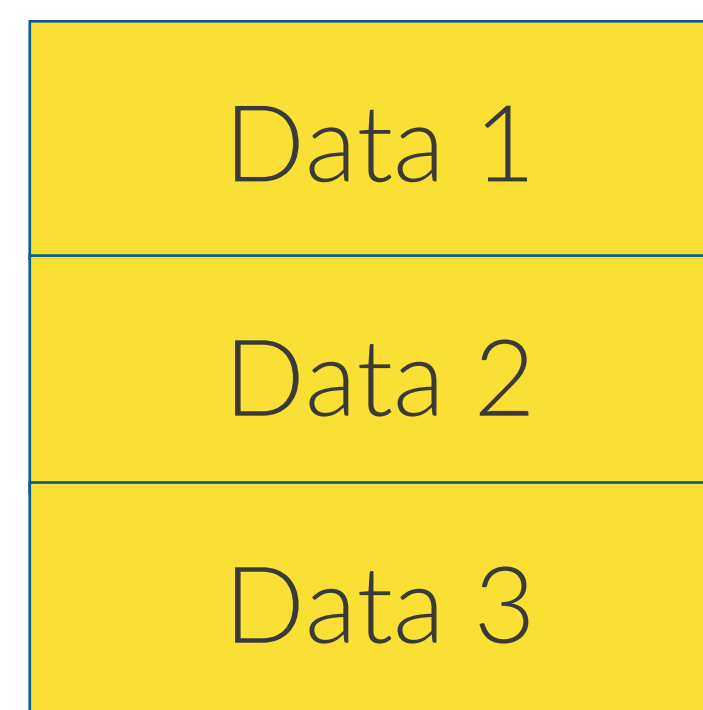
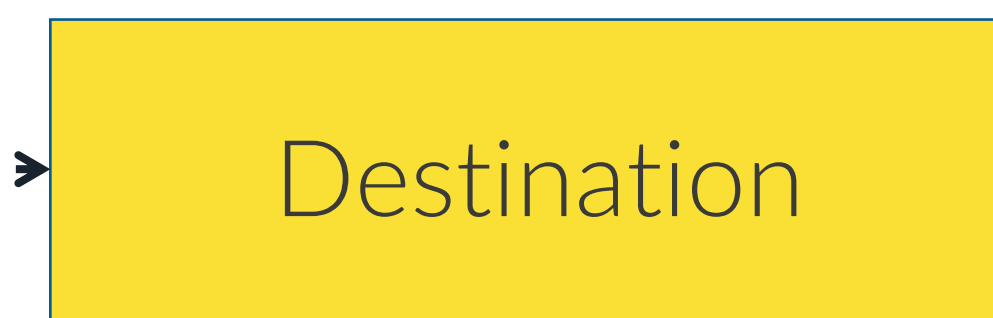
Protocol Stack : Transport Layer



Controls reliability of connection using

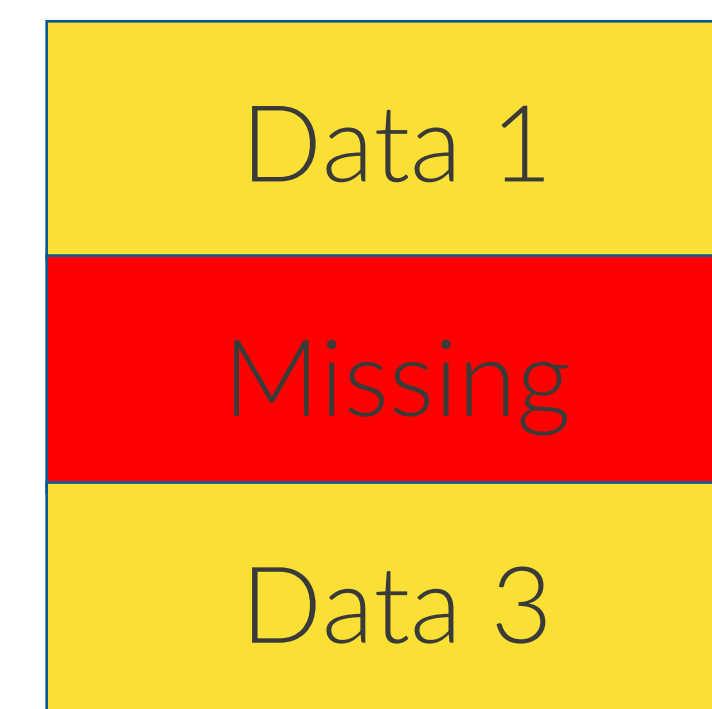
- Segmentation
- Flow Control
- Error Control

Error Control

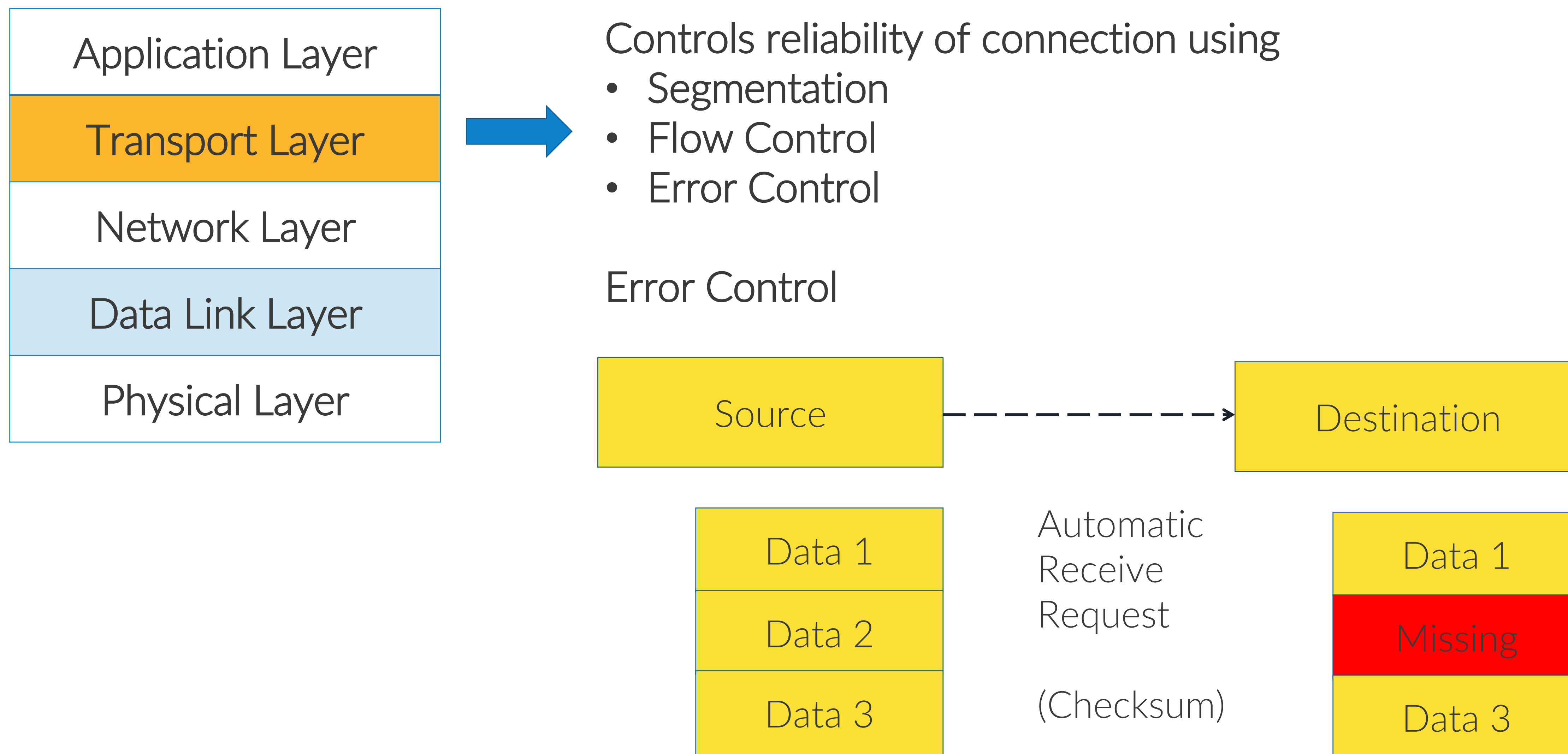


Automatic
Receive
Request

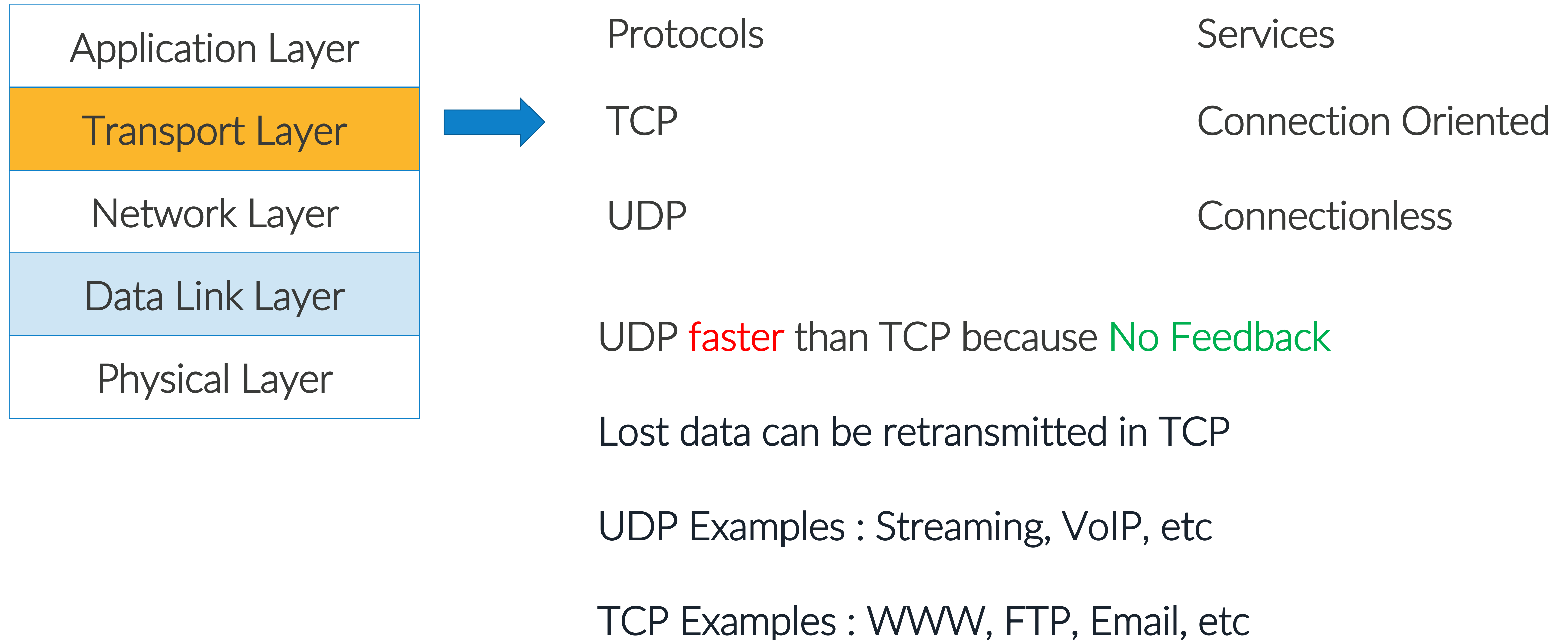
Retransmit
Data



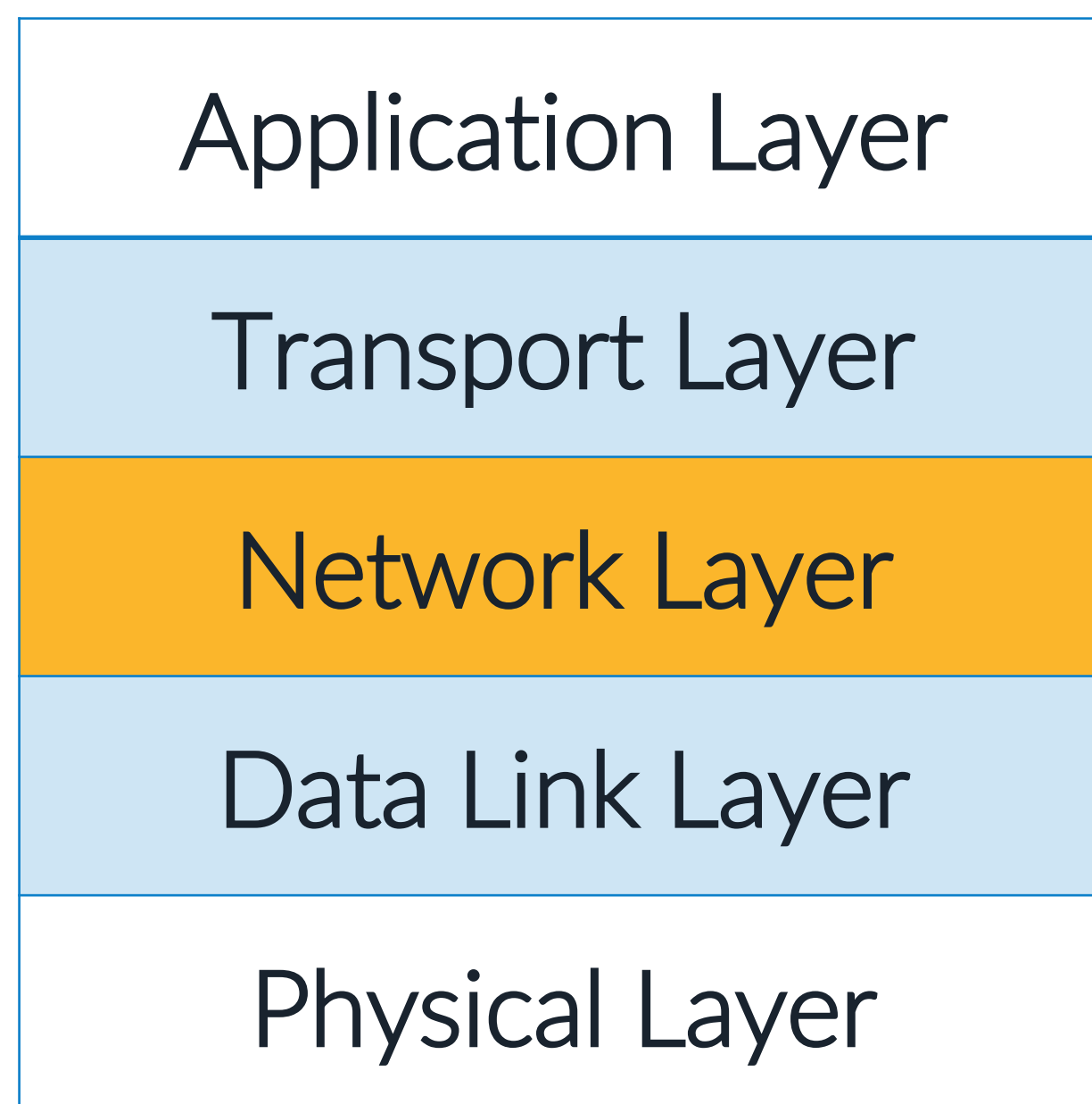
Protocol Stack : Transport Layer



Protocol Stack : Transport Layer



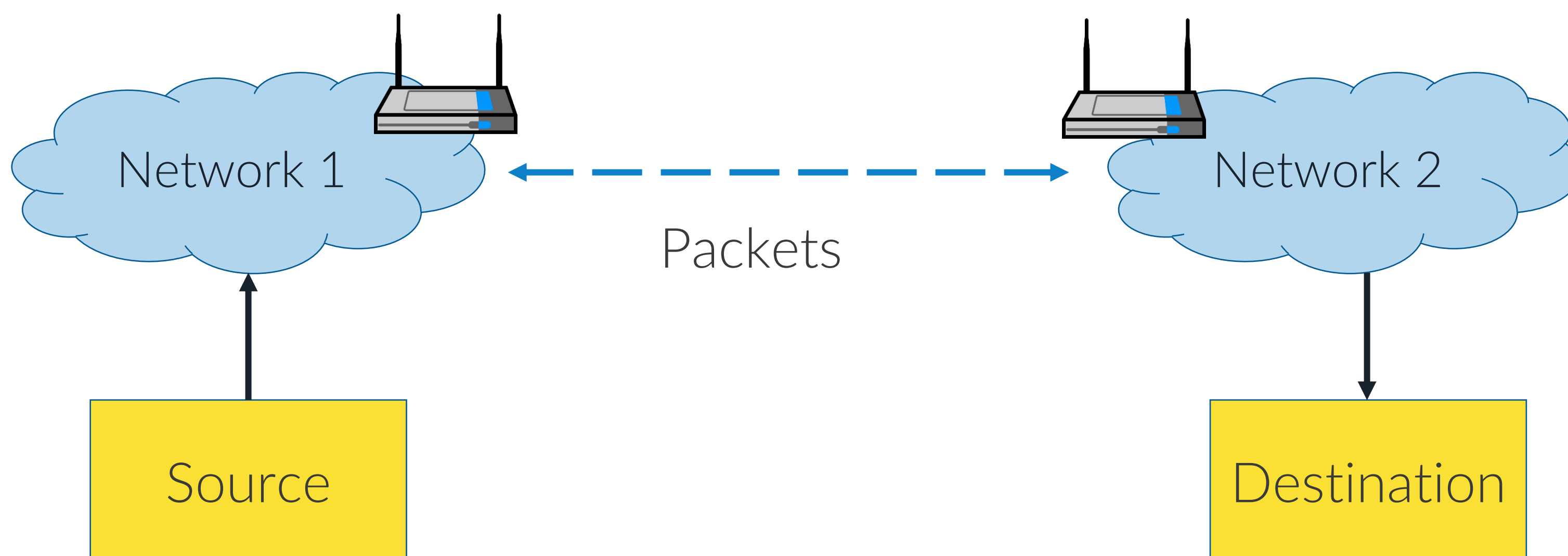
Protocol Stack : Network Layer



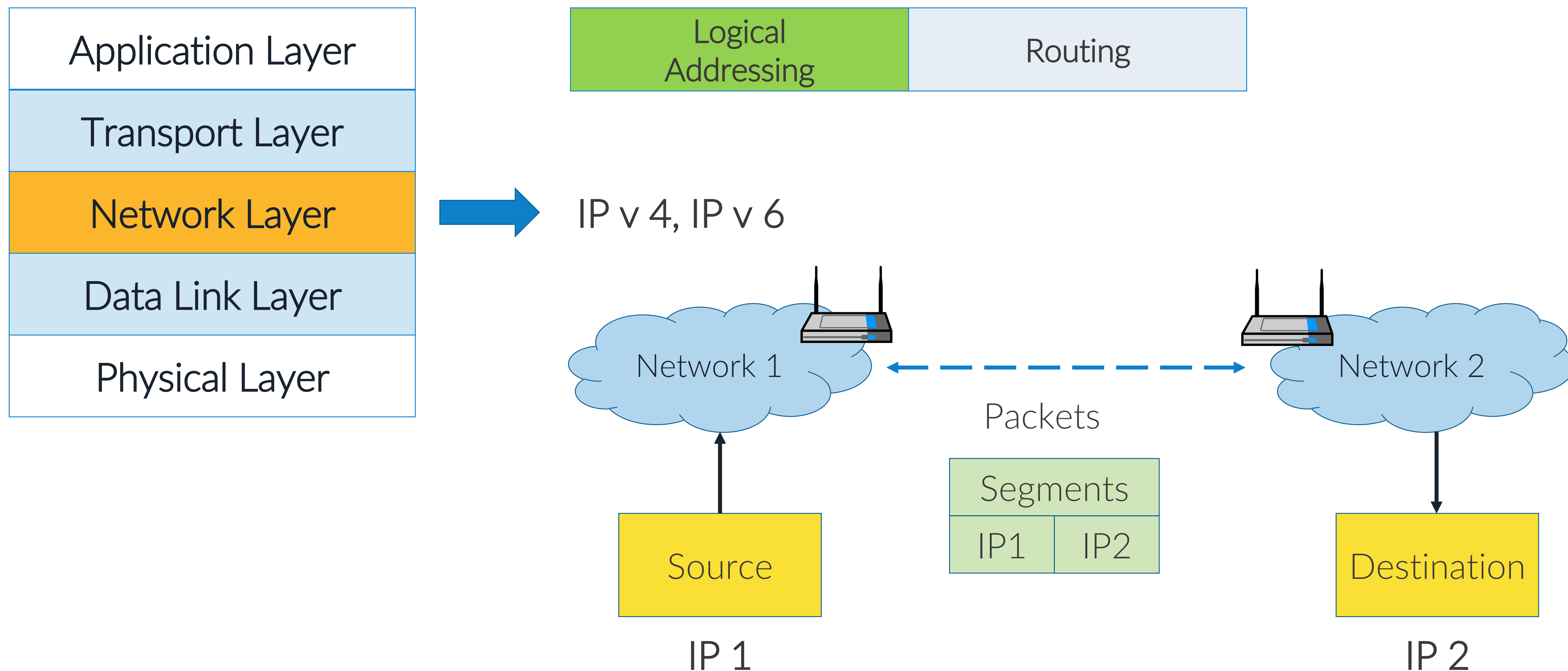
Receives **Segments** from Transport Layer

Transmission of received Data Segment from one system to other located in different networks

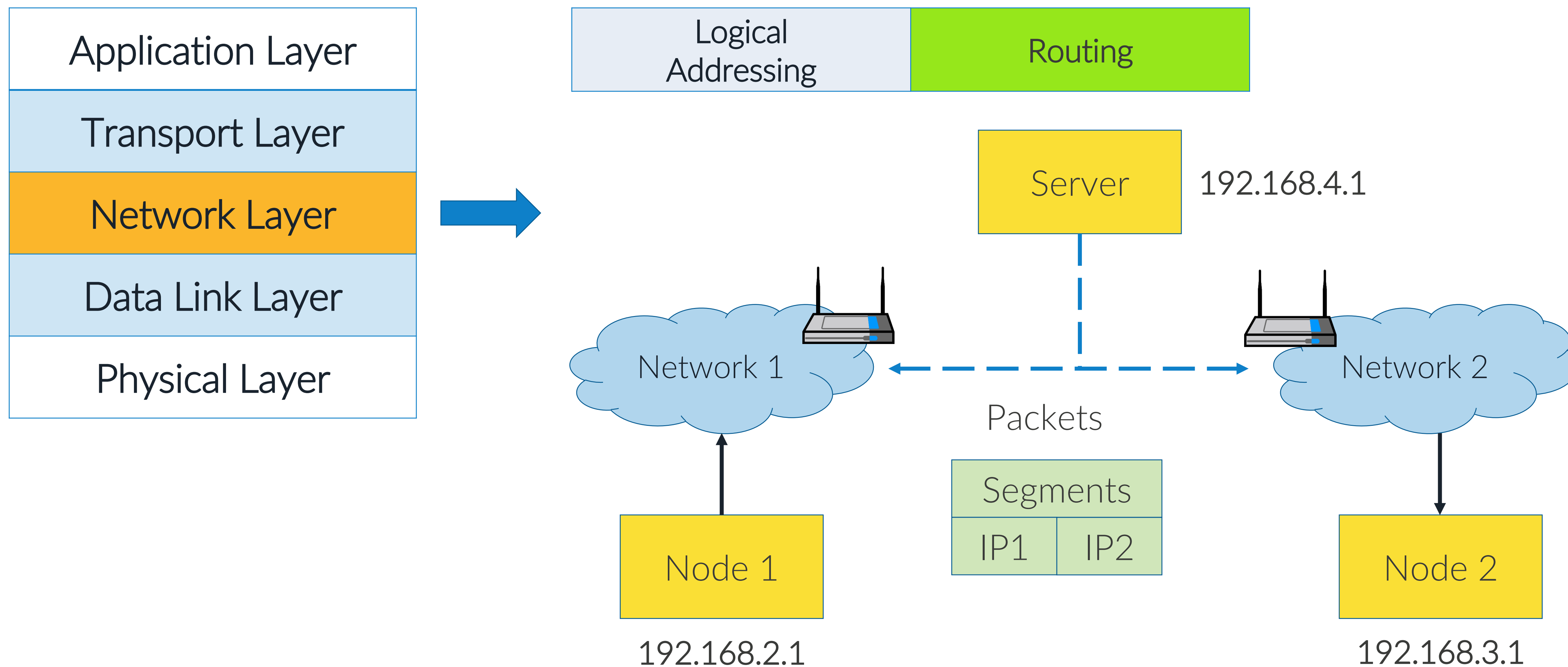
Data units called **Packets**



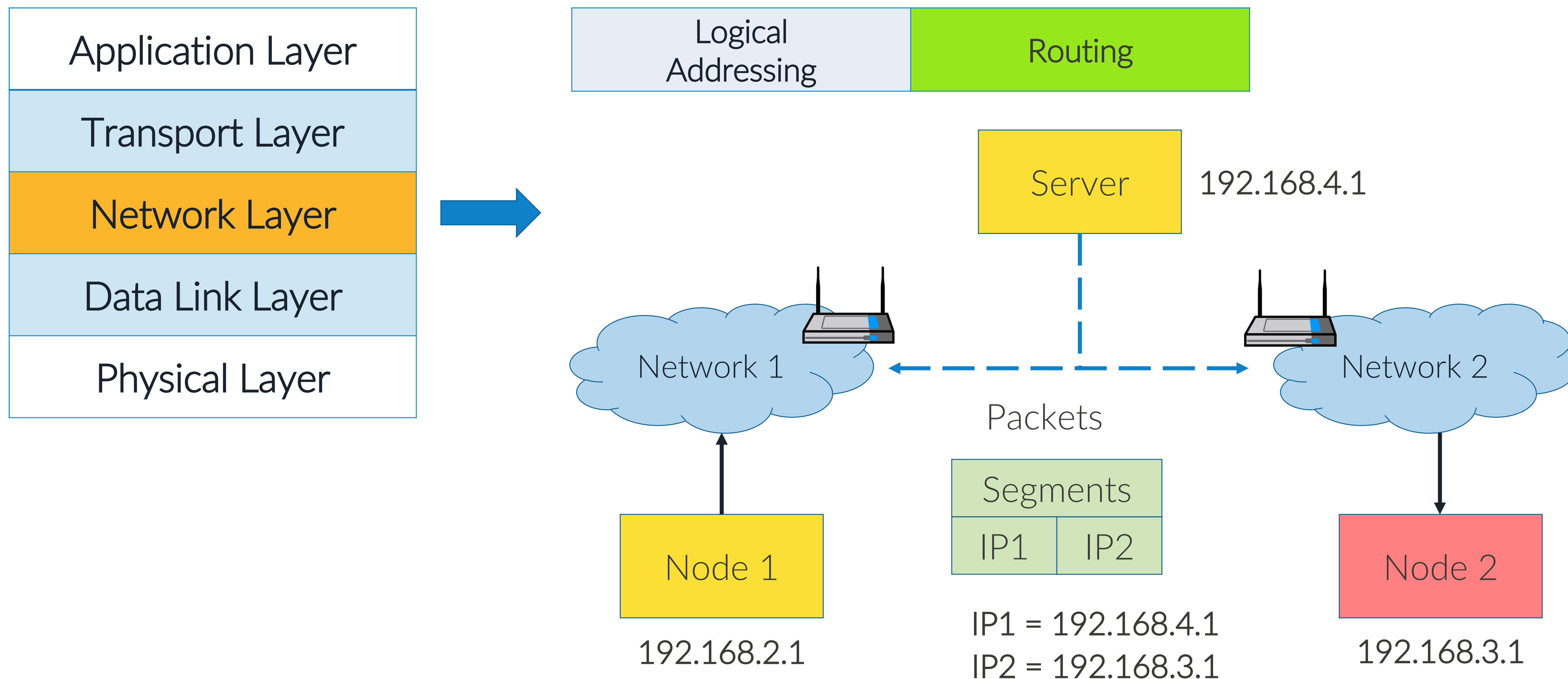
Protocol Stack : Network Layer



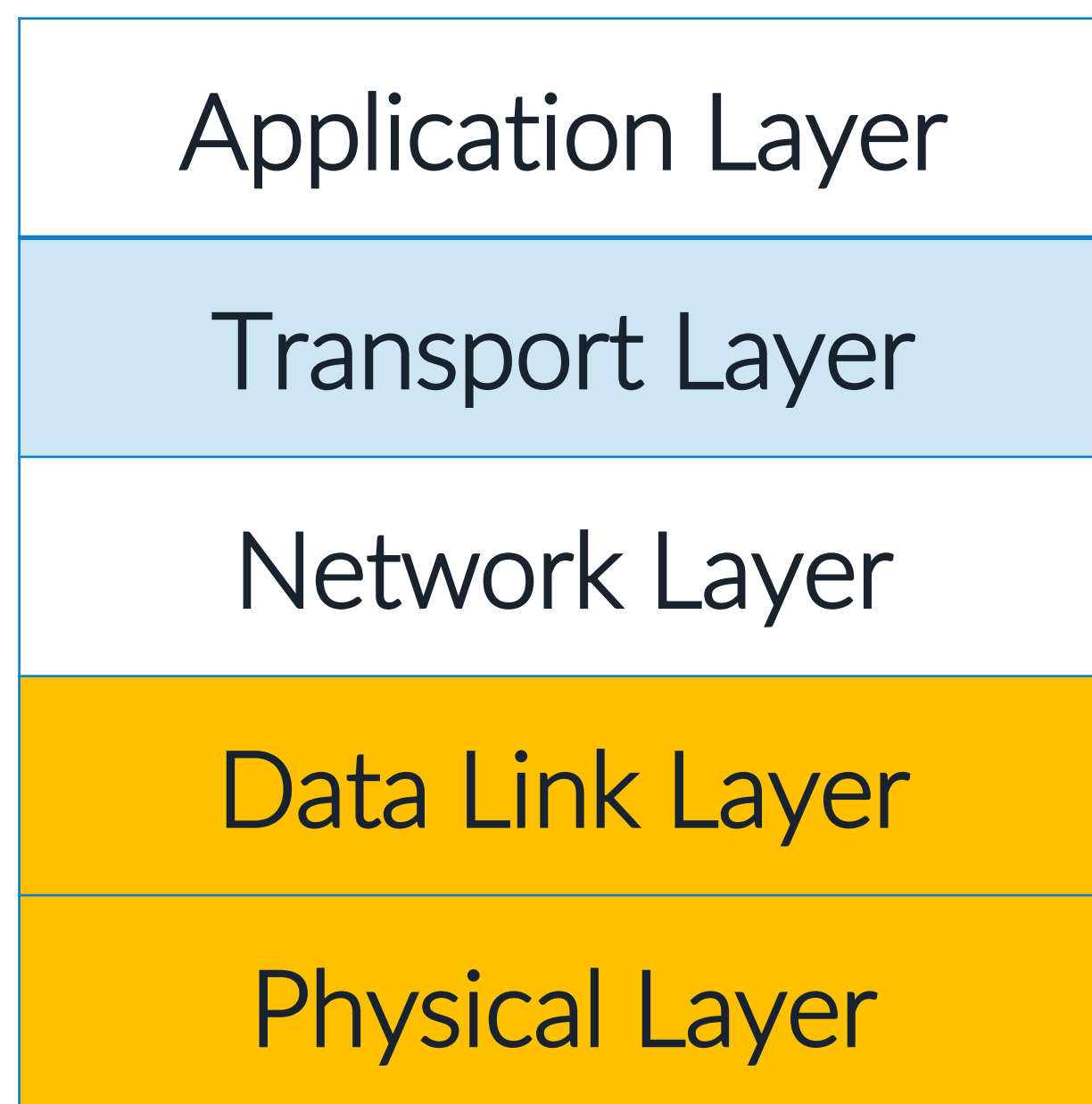
Protocol Stack : Network Layer



Protocol Stack : Network Layer



Protocol Stack : Data Link Layer



Receives **Packets** from Network Layer

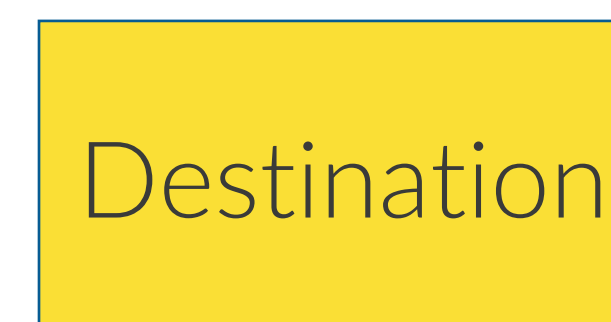
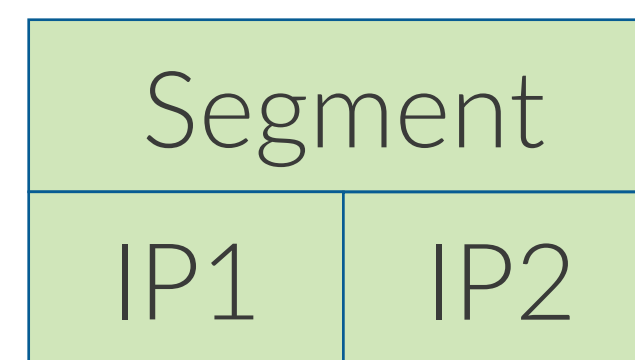
Addressing Types:

Logical : Network Layer

Physical : Data Link Layer

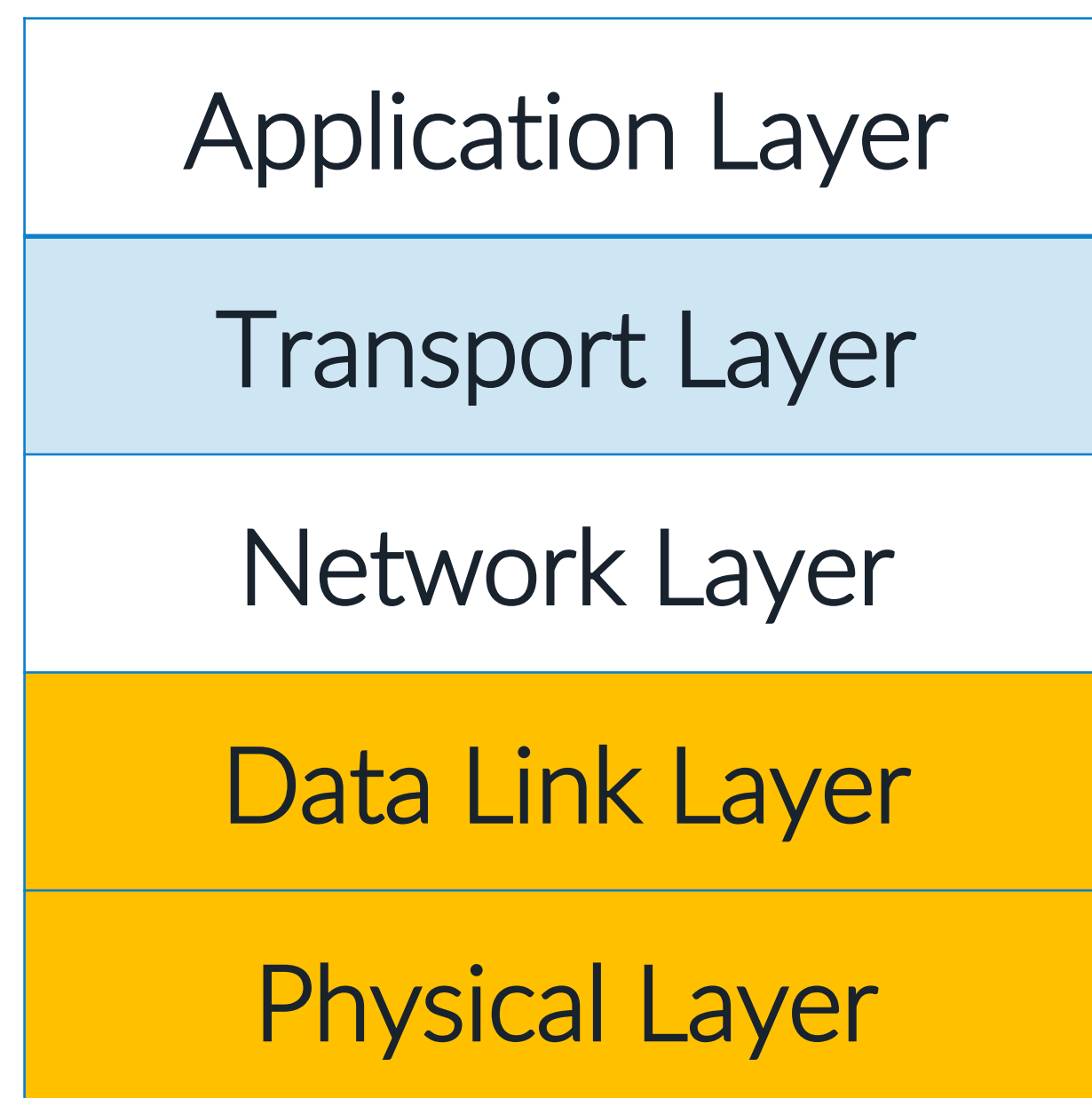


IP 1



IP 2

Protocol Stack : Data Link Layer

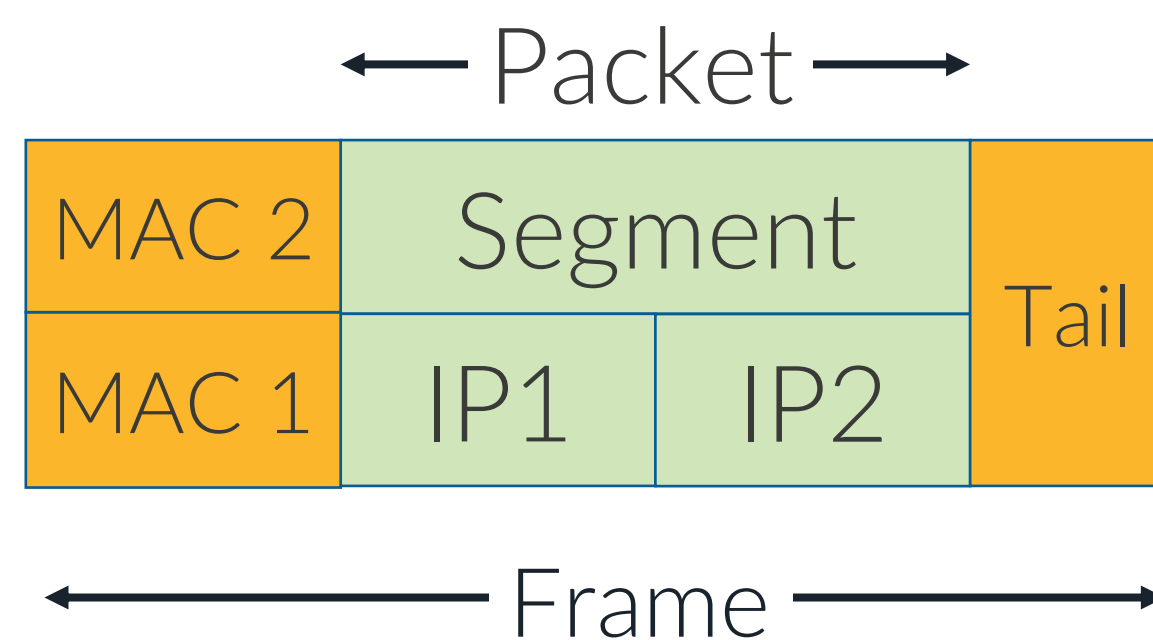
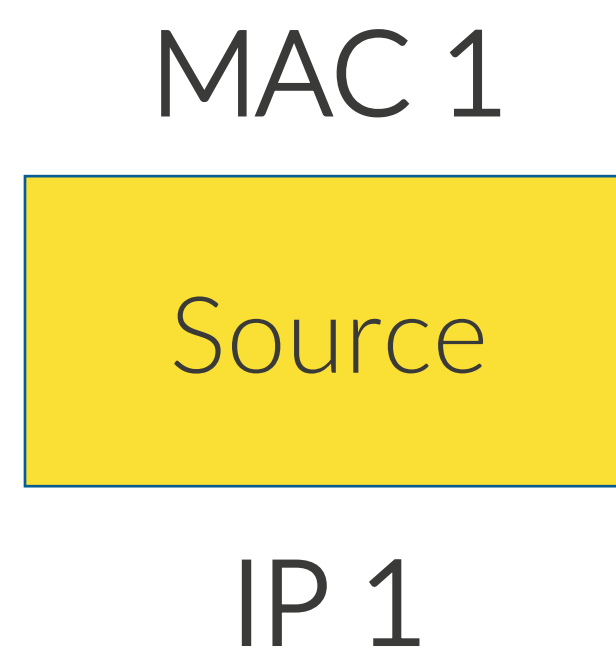


Receives **Packets** from Network Layer

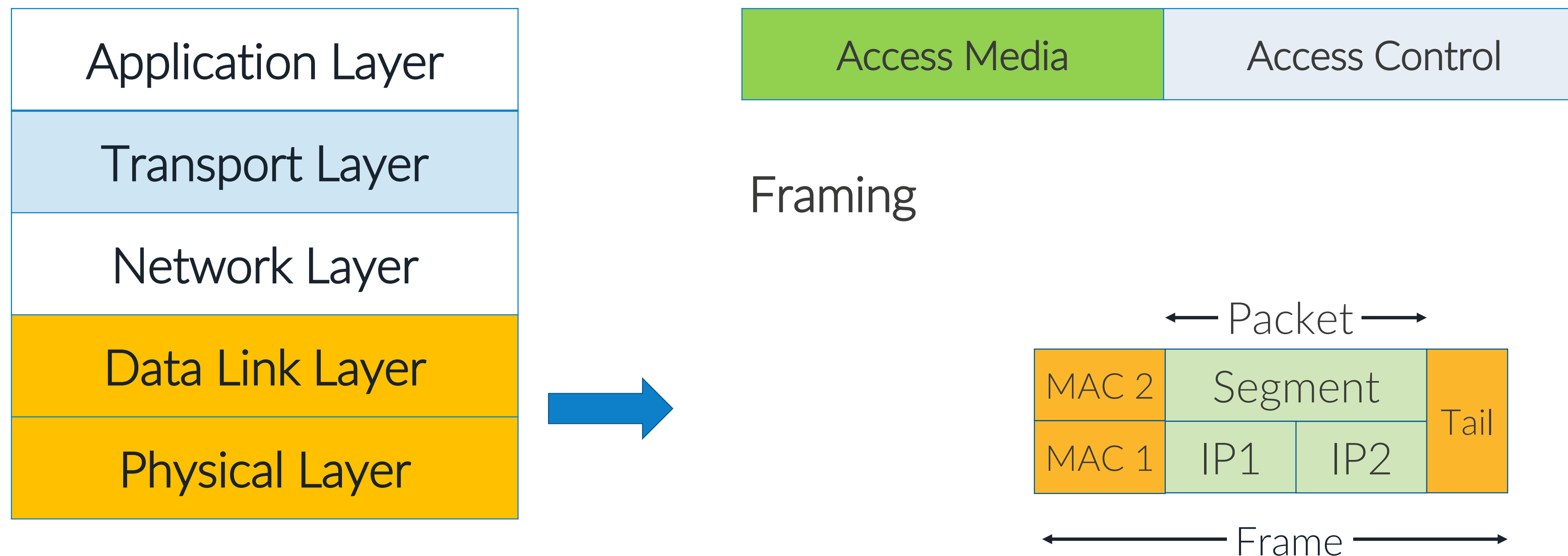
Addressing Types:

Logical : Network Layer

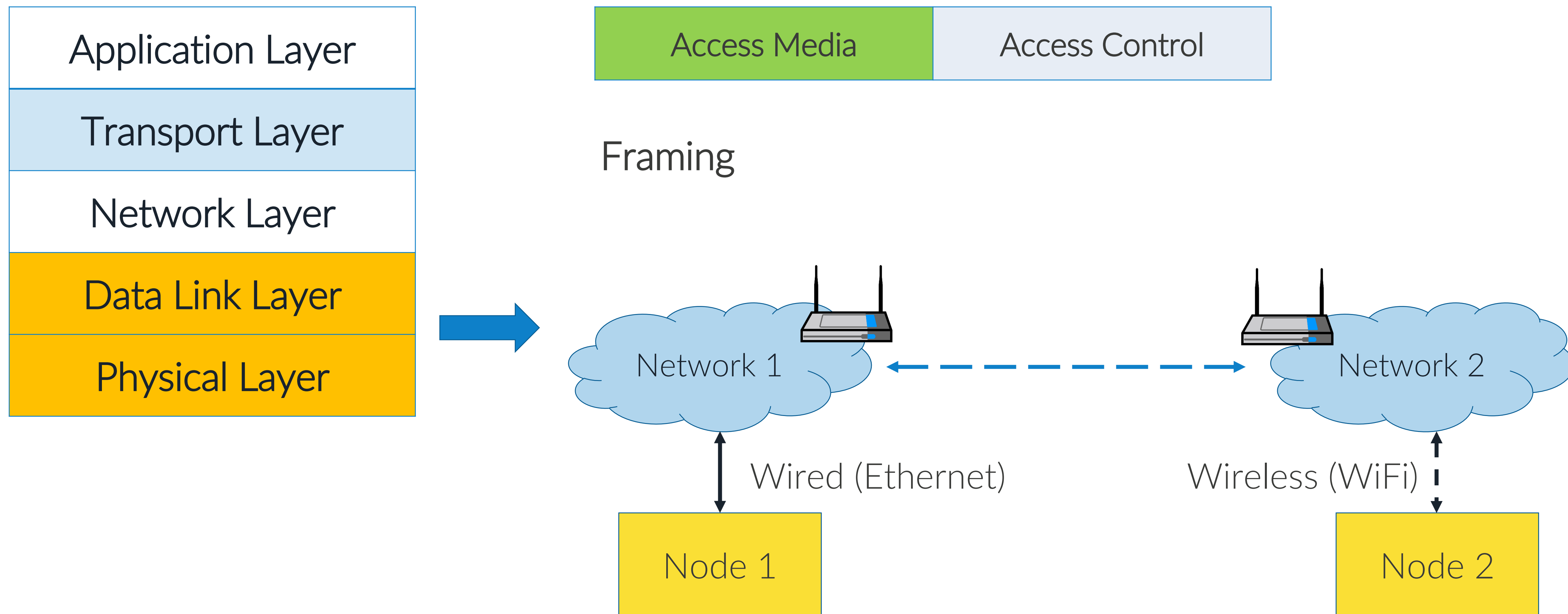
Physical : Data Link Layer



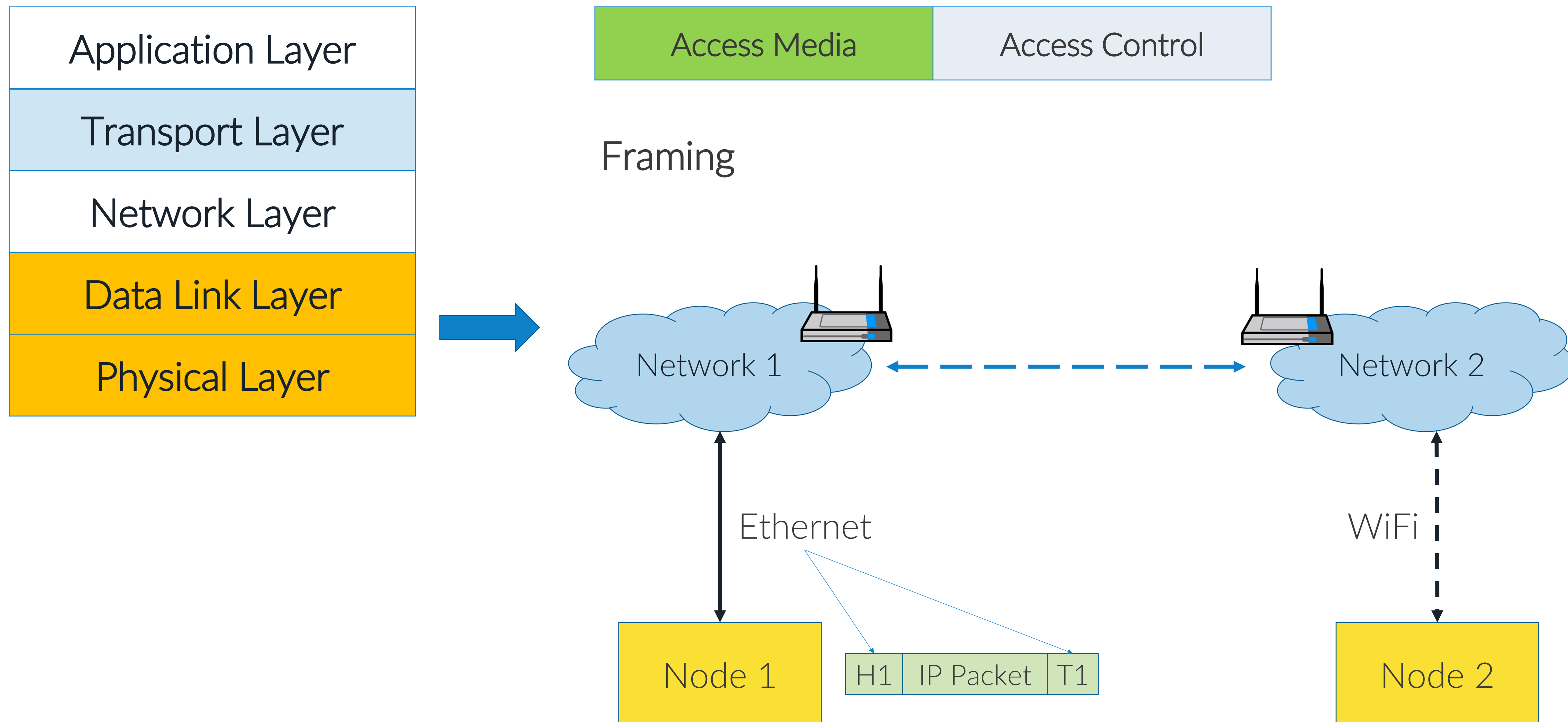
Protocol Stack : Data Link Layer



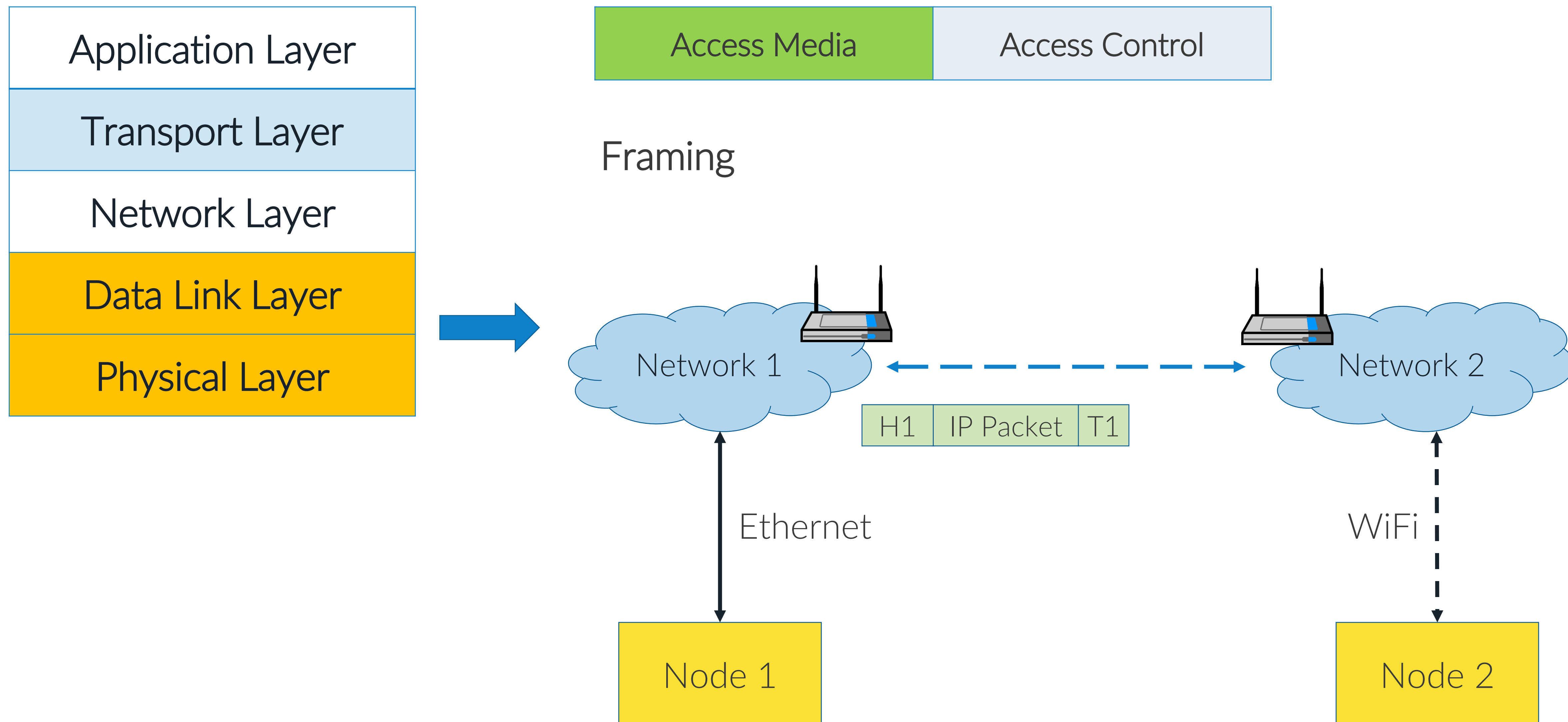
Protocol Stack : Data Link Layer



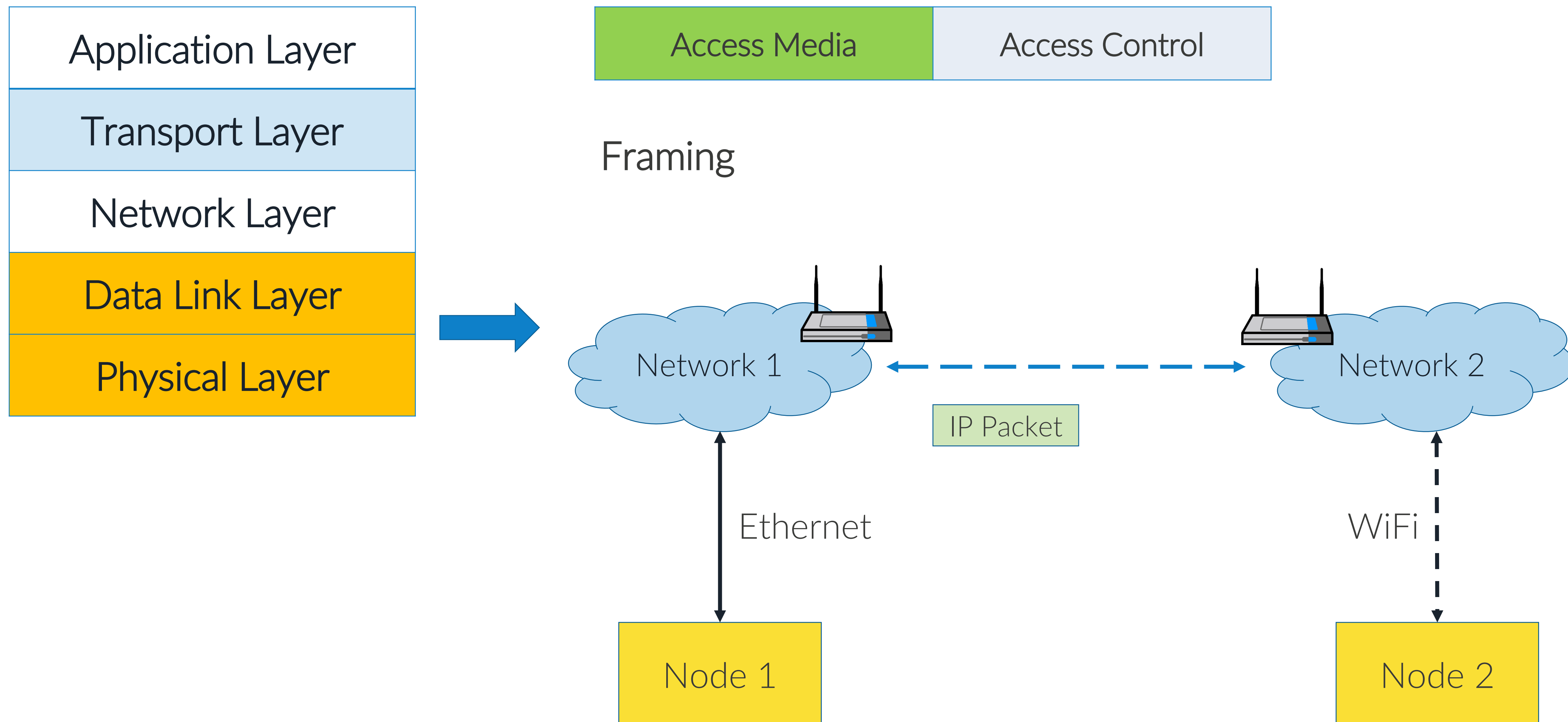
Protocol Stack : Data Link Layer



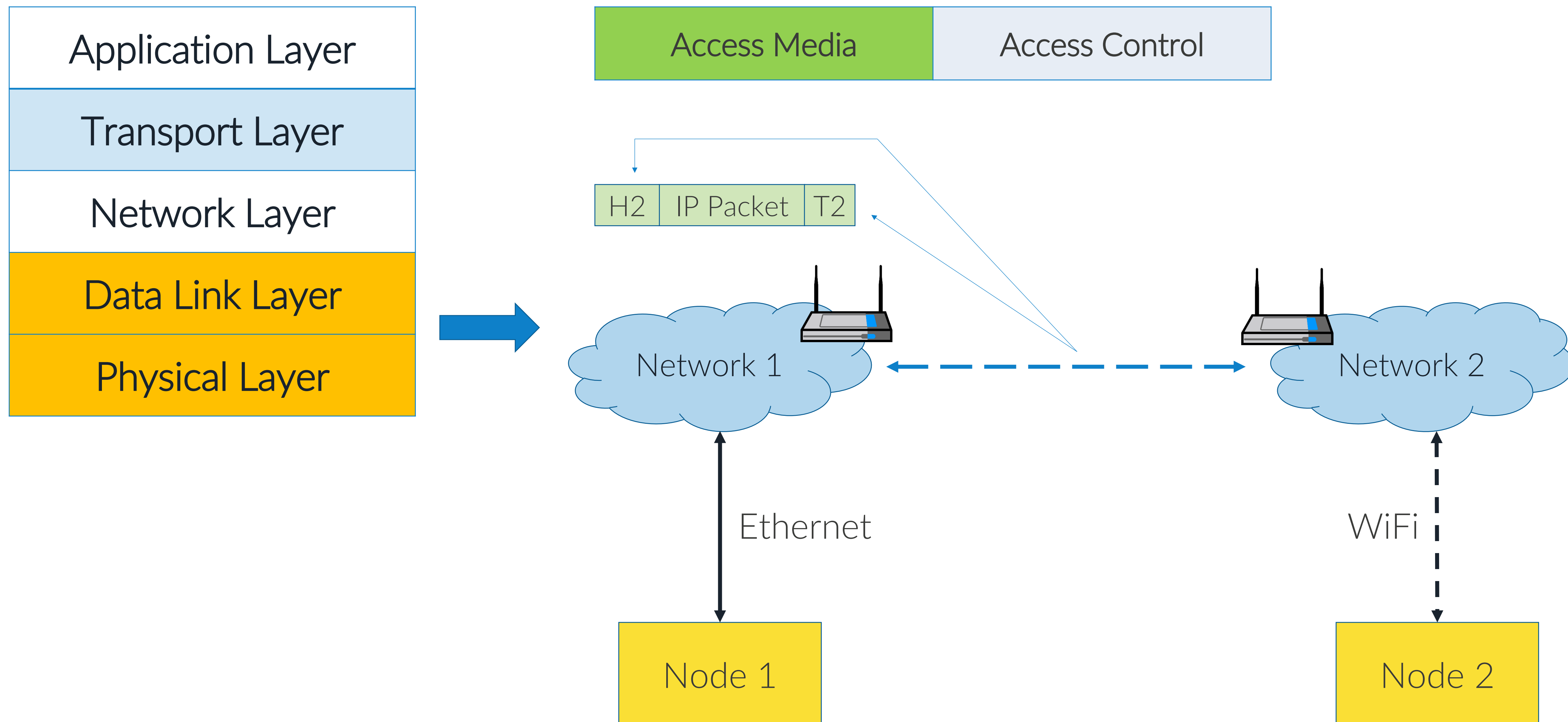
Protocol Stack : Data Link Layer



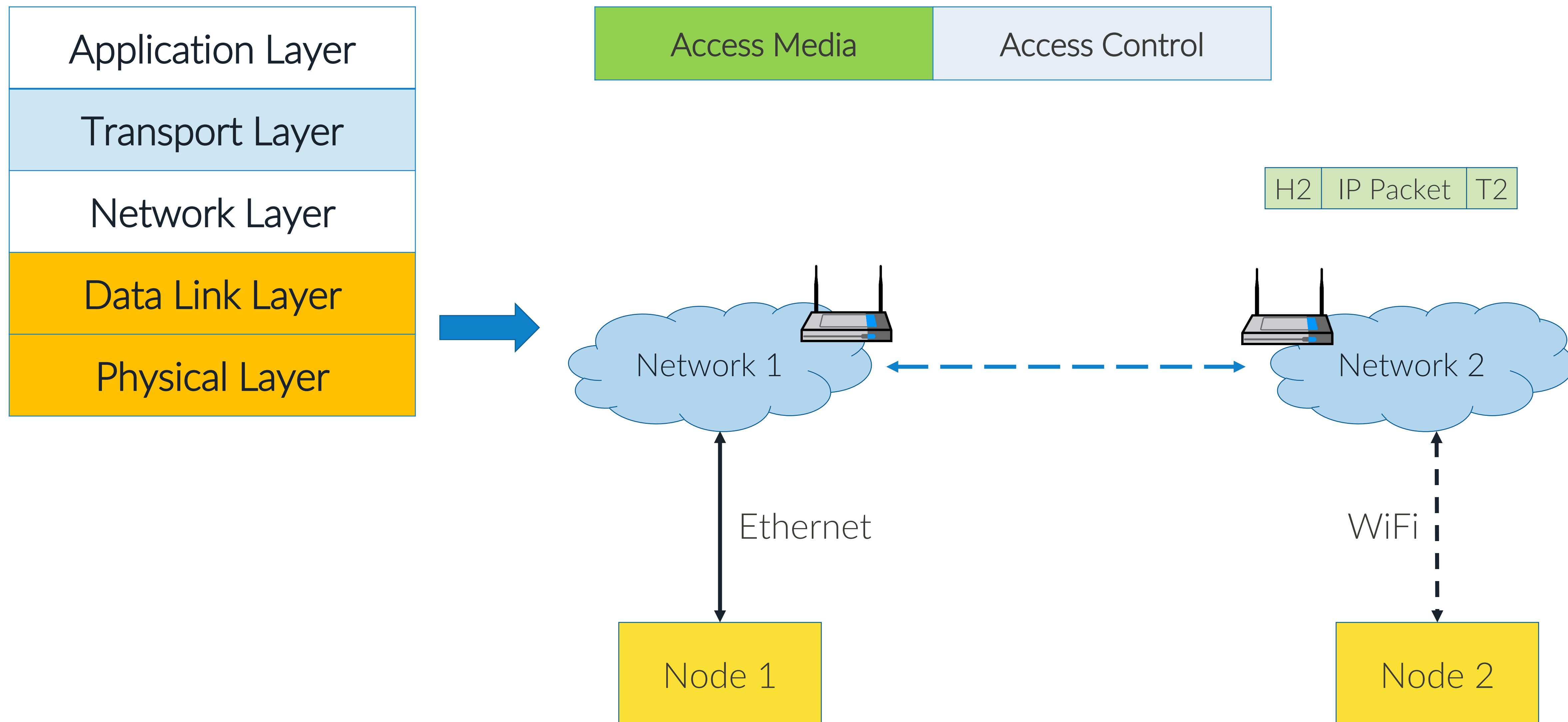
Protocol Stack : Data Link Layer



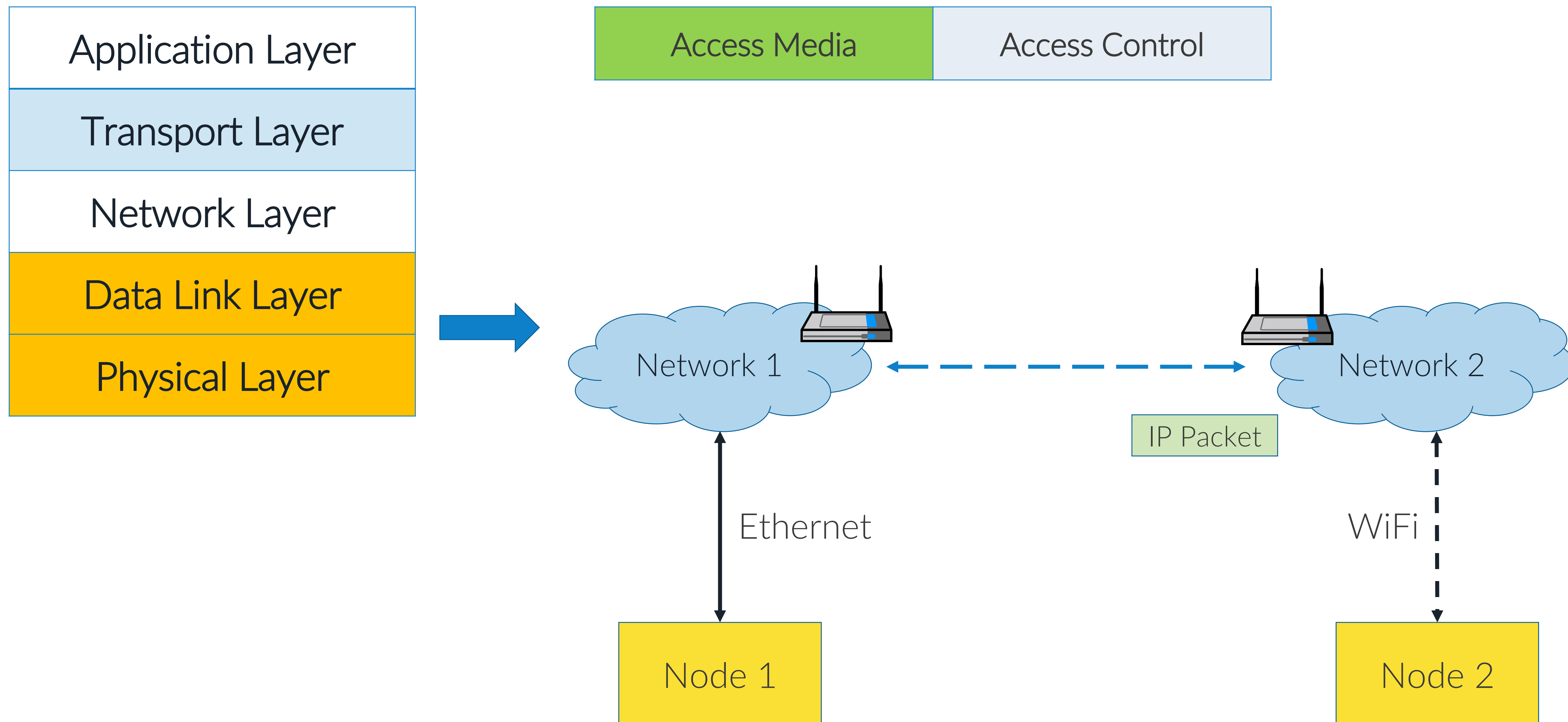
Protocol Stack : Data Link Layer



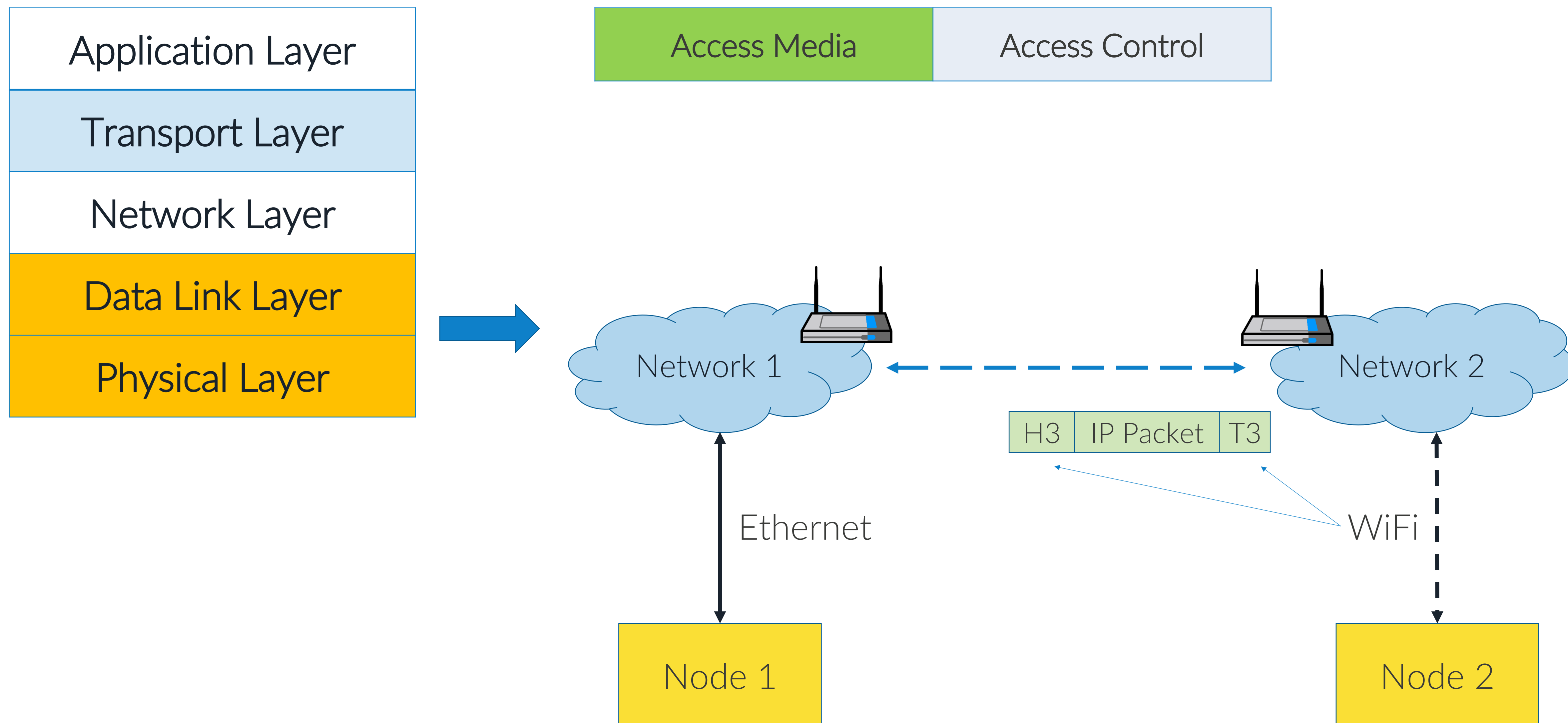
Protocol Stack : Data Link Layer



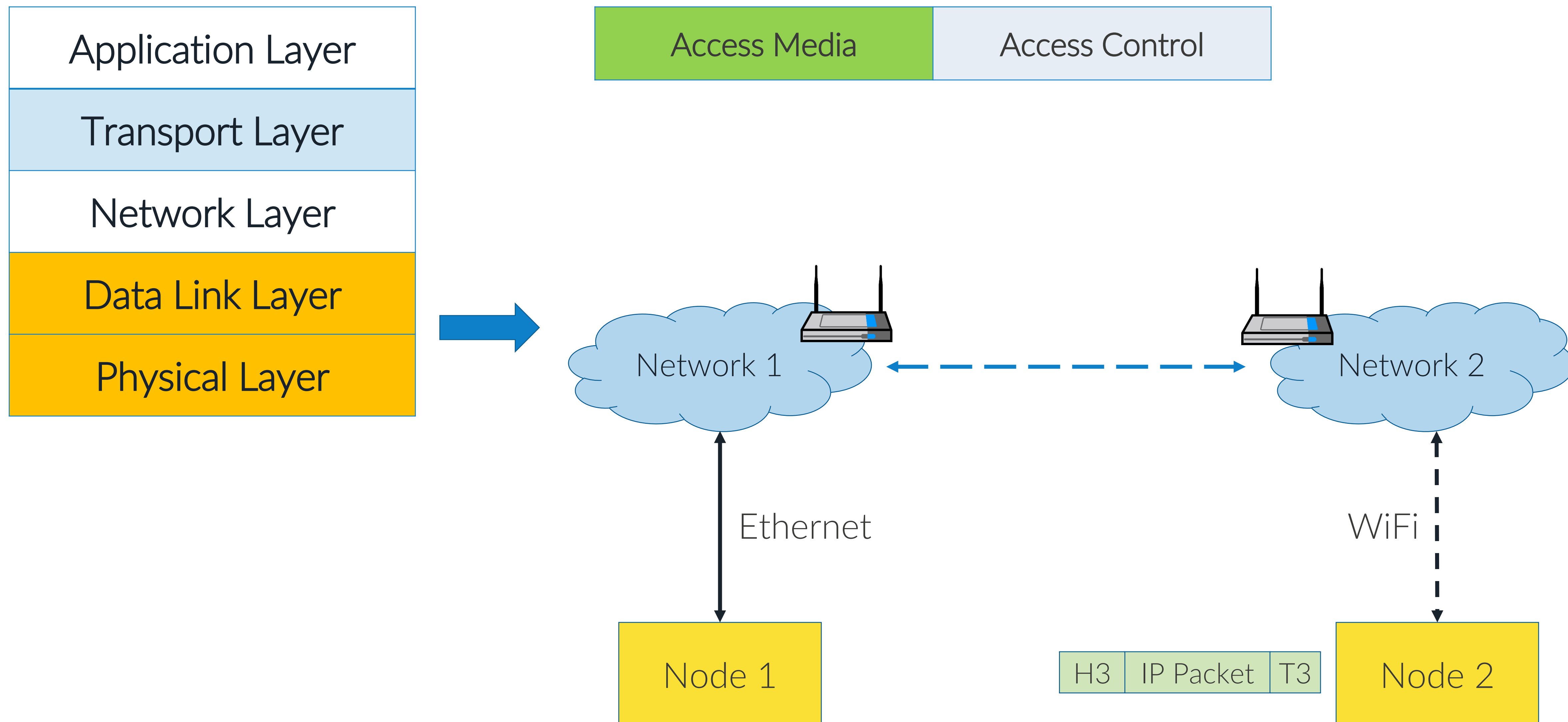
Protocol Stack : Data Link Layer



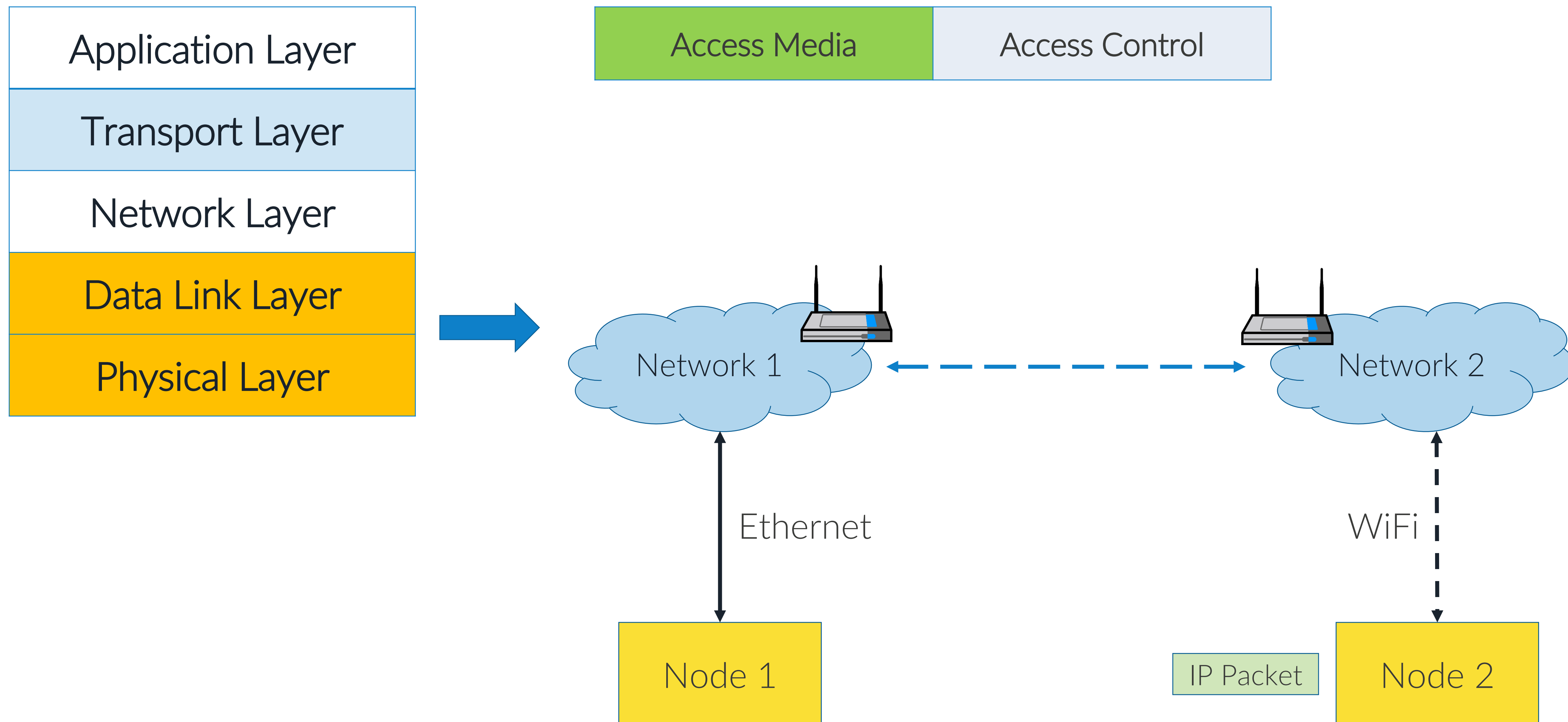
Protocol Stack : Data Link Layer



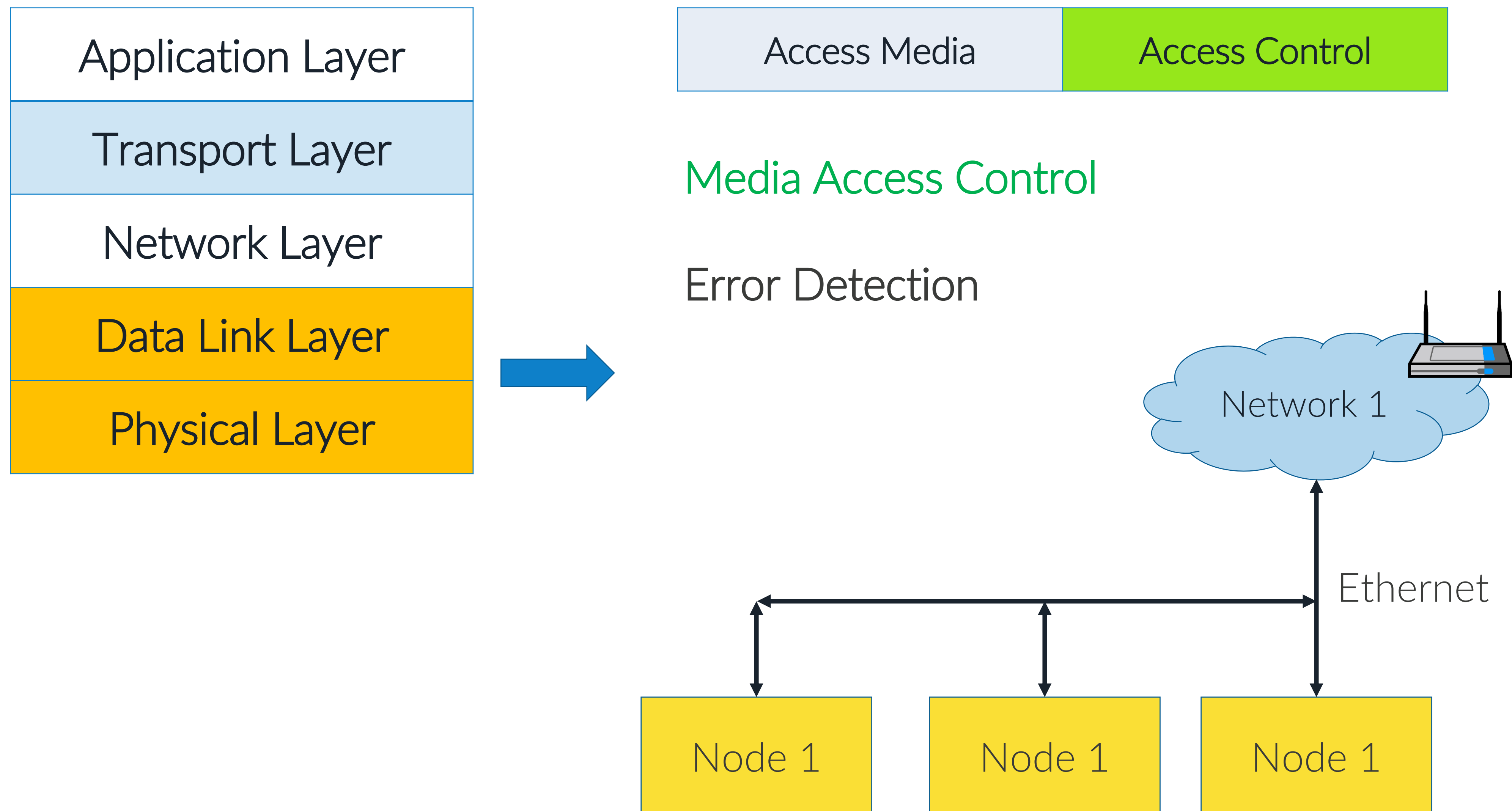
Protocol Stack : Data Link Layer



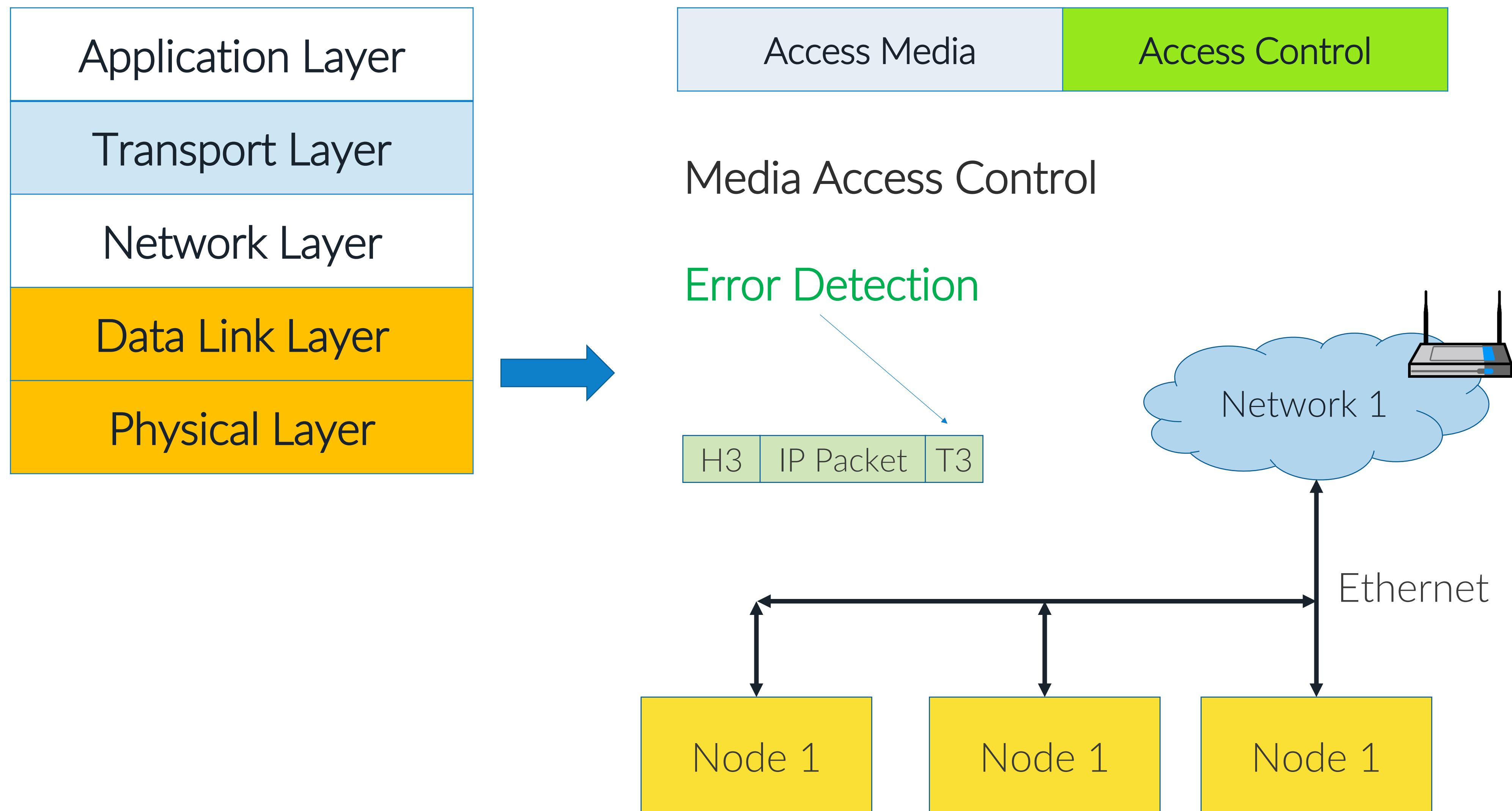
Protocol Stack : Data Link Layer



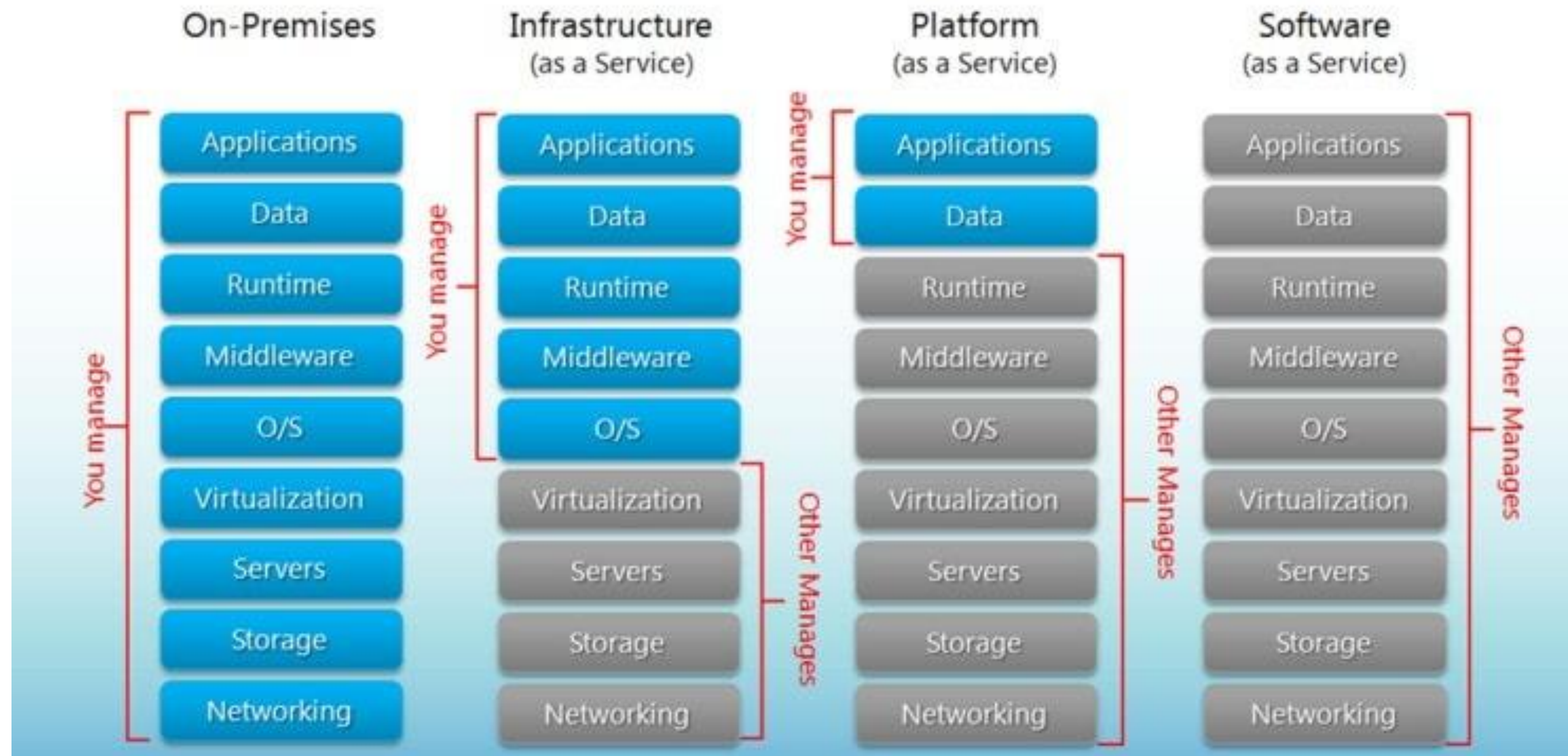
Protocol Stack : Data Link Layer



Protocol Stack : Data Link Layer



SaaS vs PaaS vs IaaS



Recap of OSI 7-Layer Model

- Physical Layer: Deals with hardware and transmission media.
- Data Link Layer: Manages node-to-node communication and error detection.
- Network Layer: Handles routing and forwarding of packets (e.g., IP).
- Transport Layer: Ensures reliable data transfer (e.g., TCP, UDP).
- Session Layer: Manages sessions between applications.
- Presentation Layer: Translates data formats (e.g., encryption, compression).
- Application Layer: Provides interfaces for end-user applications (e.g., HTTP, FTP).

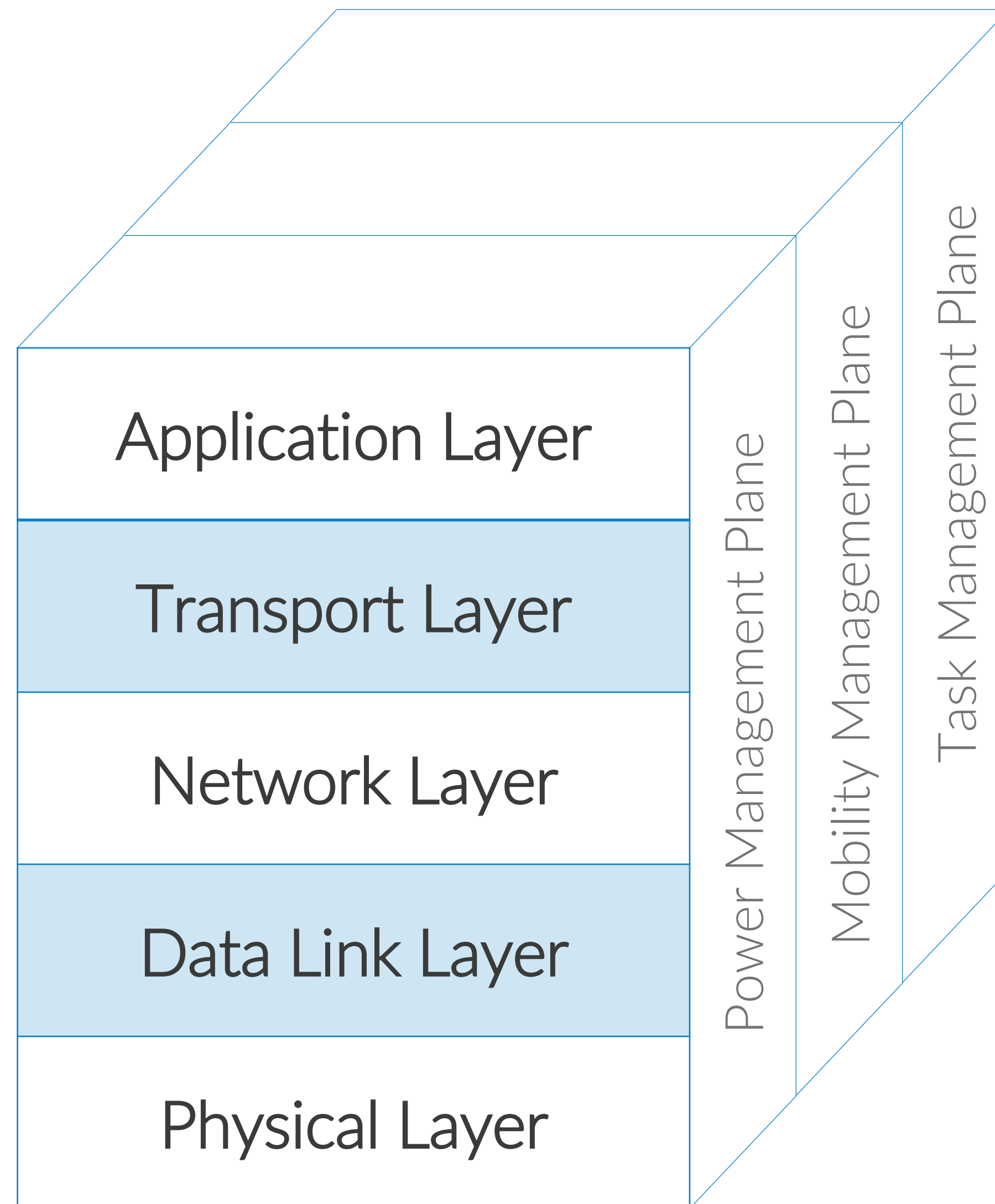
Challenges of IoT vs Traditional Networking

- Resource Constraints :
 - Low power, limited memory, and processing capabilities.
- Heterogeneity :
 - Diverse devices with varying requirements (e.g., sensors, actuators, gateways).
- Scalability :
 - Millions of connected devices in IoT networks.
- Low Power and Low Bandwidth :
 - Energy efficiency is critical for battery-powered devices.
- Mobility and Interoperability :
 - Devices may move across networks or use different standards.

Challenges of IoT vs Traditional Networking

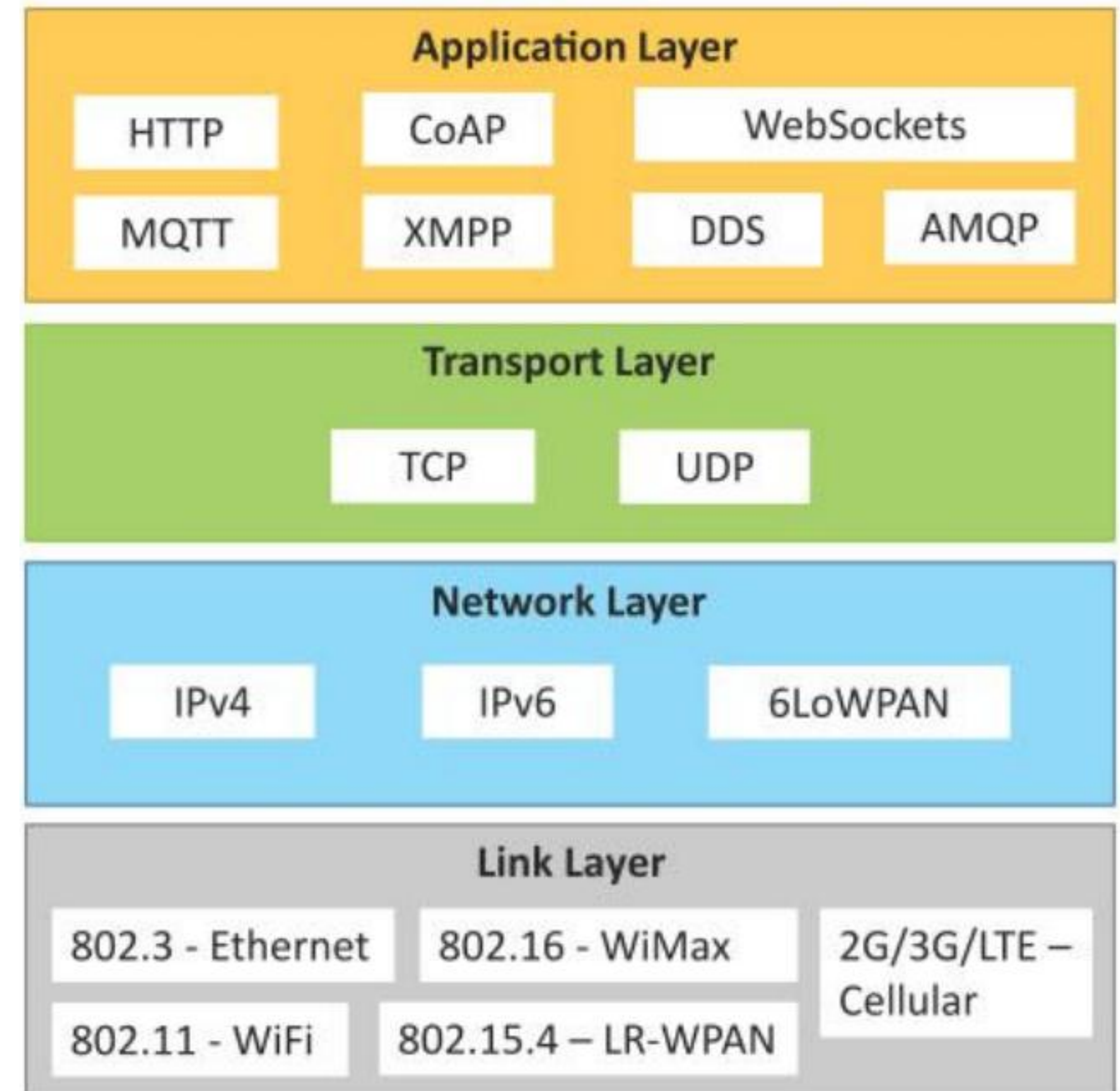
Device Resources	Devices typically have ample power, memory, and processing capabilities.	Devices are resource-constrained (low power, limited memory, and processing power).
Network Scale	Designed for smaller-scale networks (e.g., LANs, WANs).	Designed for massive scalability (millions or billions of connected devices).
Power Consumption	Power consumption is not a primary concern (devices are often plugged in).	Energy efficiency is critical (battery-powered devices must last months/years).
Bandwidth	High bandwidth is available (e.g., Ethernet, Wi-Fi).	Low bandwidth is typical (e.g., LoRaWAN, Zigbee, NB-IoT).
Communication Range	Short-range communication is common (e.g., Wi-Fi, Ethernet).	Long-range communication is often required (e.g., LoRaWAN, Sigfox, NB-IoT).
Protocols	Protocols like TCP/IP, HTTP, FTP are standard.	Lightweight protocols like MQTT, CoAP, and 6LoWPAN are used.
Data Size	Large data packets are common (e.g., video streaming, file transfers).	Small data packets are typical (e.g., sensor readings, status updates).
Mobility	Mobility is limited (e.g., laptops, smartphones).	High mobility is common (e.g., wearable devices, vehicles, drones).
Interoperability	Interoperability is less complex (fewer device types).	Interoperability is critical (heterogeneous devices with different standards).
Latency	Latency is acceptable for most applications (e.g., web browsing, email).	Ultra-low latency is required for real-time applications (e.g., industrial IoT).
Security	Security is important but not always prioritized (e.g., firewalls, encryption).	Security is paramount due to the sensitive nature of IoT data (e.g., health, safety).
Topology	Star or mesh topologies are common (e.g., Wi-Fi, Ethernet).	Mesh, star, and hybrid topologies are used (e.g., Zigbee, LoRaWAN).
Deployment Environment	Typically indoor or controlled environments (e.g., offices, homes).	Often deployed in harsh or remote environments (e.g., agriculture, smart cities).
Data Processing	Data is processed centrally (e.g., cloud servers).	Edge computing is common (data processed locally on devices or gateways).
Cost	Cost is less of a concern (infrastructure is well-established).	Cost is critical (low-cost devices and protocols are essential).

Protocol Stack : Wireless Sensor Networks



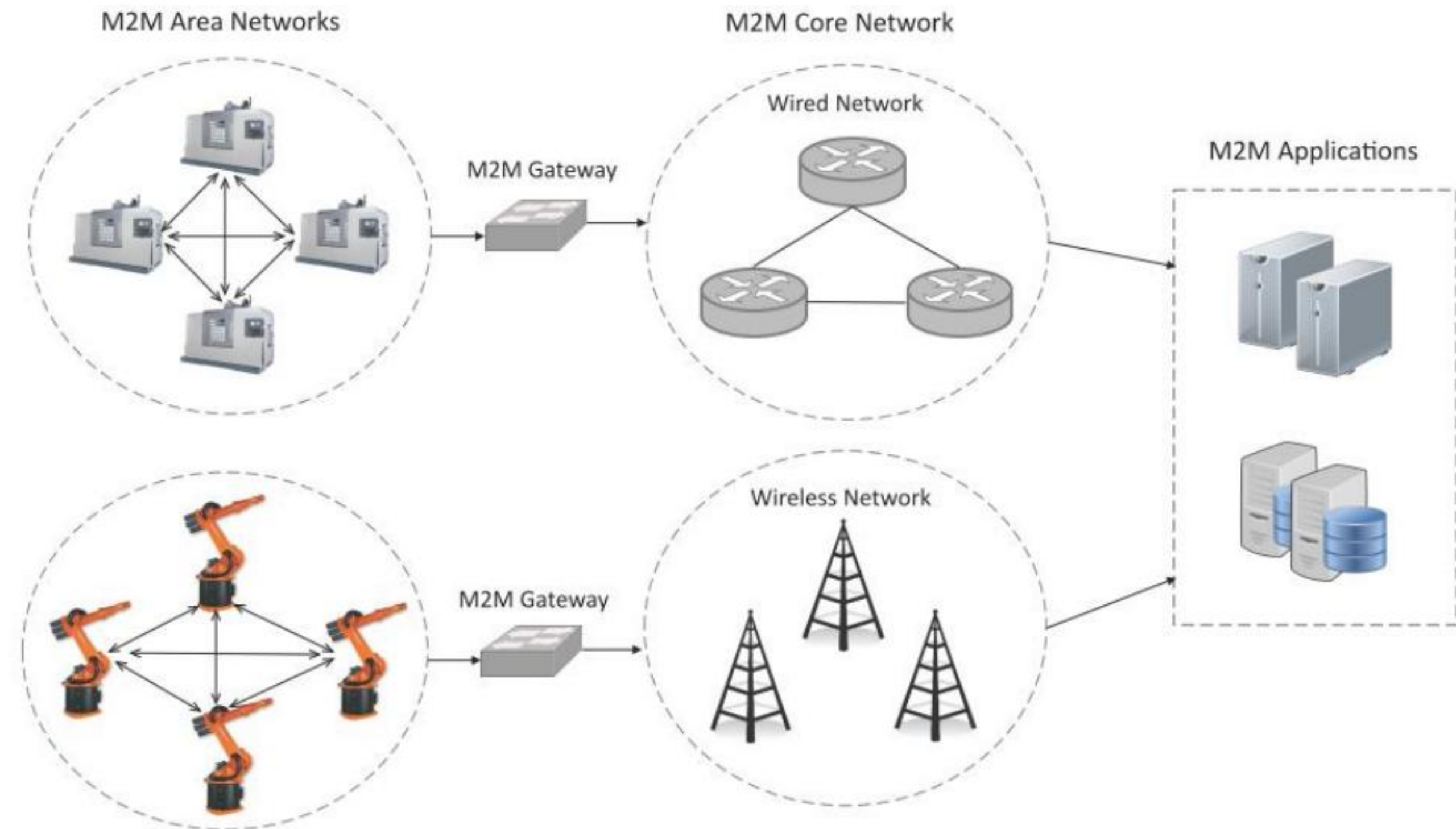
IoT Protocol Stack

- **Link Layer**
 - 802.3 – Ethernet
 - 802.11 – WiFi
 - 802.16 – WiMax
 - 802.15.4 – ZigBee
 - 2G/3G/4G
- **Network Layer**
 - IPv4
 - IPv6
 - 6LoWPAN
- **Transport Layer**
 - TCP
 - UDP
- **Application Layer**
 - HTTP
 - CoAP
 - WebSocket
 - MQTT
 - XMPP
 - DDS
 - AMQP



Machine to Machine communication

- Machine-to-Machine (M2M) refers to networking of machines (or devices) for the purpose of remote monitoring and control and data exchange



Machine to Machine communication

- An M2M area network comprises of machines (or M2M nodes) which have embedded hardware modules for sensing, actuation and communication
- Various communication protocols can be used for M2M local area networks such as ZigBee, Bluetooth, ModBus, M-Bus, Wireless M-Bus, Power Line Communication (PLC), 6LoWPAN, IEEE 802.15.4, etc.

Machine to Machine communication

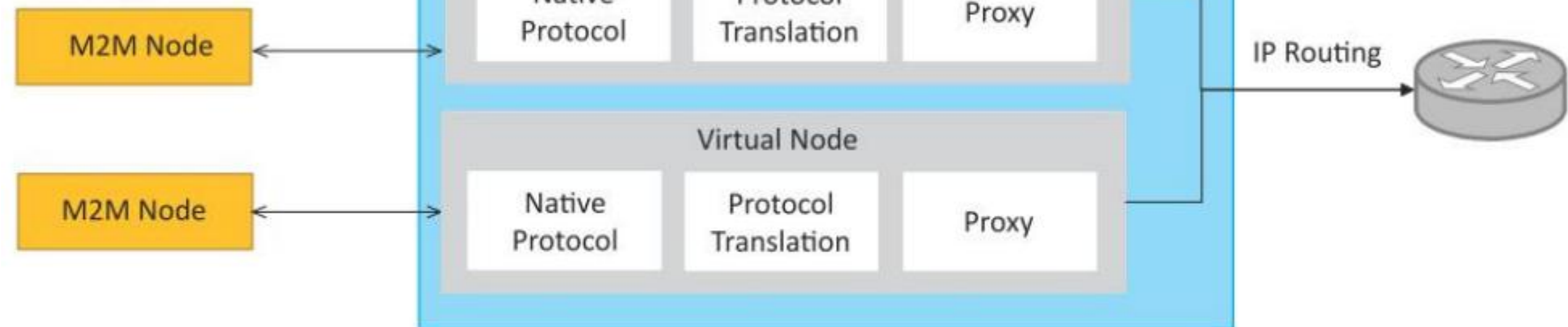
- The communication network provides connectivity to remote M2M area networks
- The communication network can use either wired or wireless networks (IP- based)
- While the M2M area networks use either proprietary or non-IP based communication protocols, the communication network uses IP-based networks

M2M Gateway

- Since non-IP based protocols are used within M2M area networks, the M2M nodes within one network cannot communicate with nodes in an external network
- To enable the communication between remote M2M area networks, M2M gateways are used

M2M Area Networks:

- Bluetooth
- ZigBee
- 802.15.4
- 6LoWPAN
- M-Bus, Wireless M-Bus
- UWB
- ModBus
- Z-Wave



Difference between M2M and IoT

- **Communication Protocols**
- M2M and IoT can differ in how the communication between the machines or devices happens
- M2M uses either proprietary or non-IP based communication protocols for communication within the M2M area networks

Difference between M2M and IoT

- **Machines in M2M vs Things in IoT**
- The "Things" in IoT refers to physical objects that have unique identifiers and can sense and communicate with their external environment (and user applications) or their internal physical states
- M2M systems, in contrast to IoT, typically have homogeneous machine types within an M2M area network

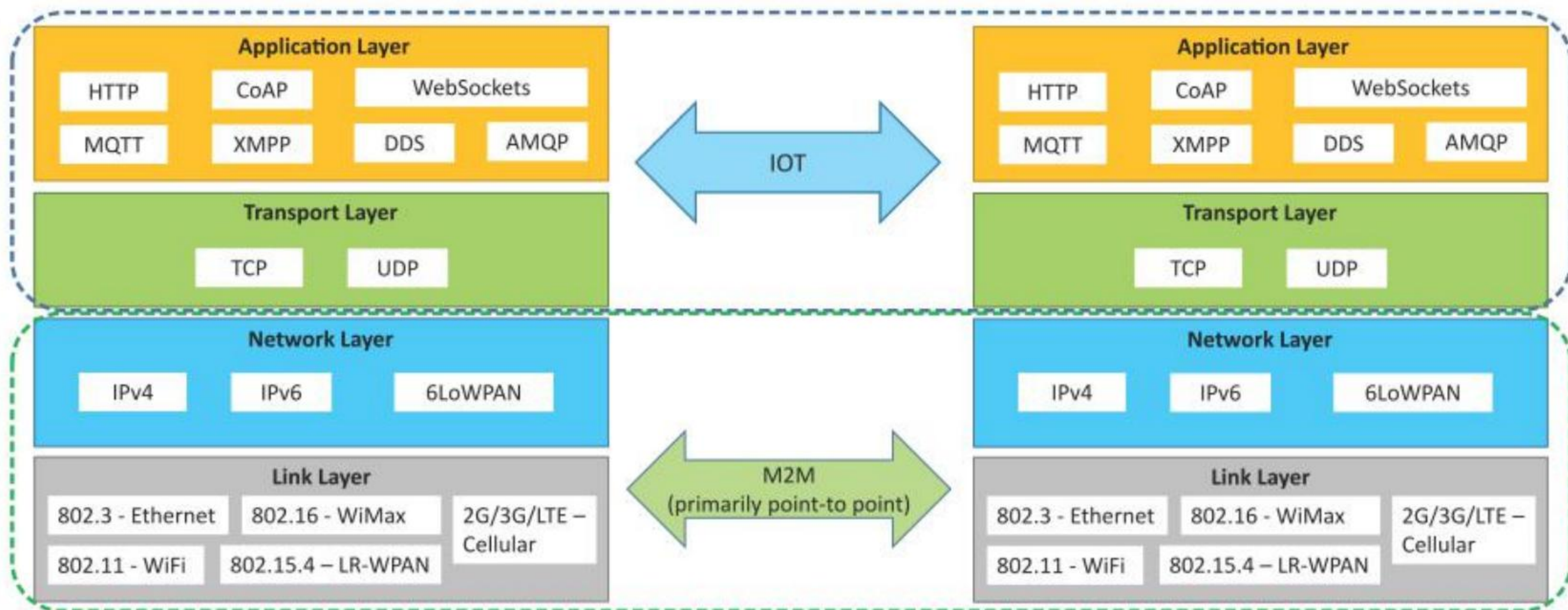
Difference between M2M and IoT

- **Hardware vs Software Emphasis**
- While the emphasis of M2M is more on hardware with embedded modules, the emphasis of IoT is more on software
- **Data Collection & Analysis**
- M2M data is collected in point solutions and often in on-premises storage infrastructure
- In contrast to M2M, the data in IoT is collected in the cloud (can be public, private or hybrid cloud)

Difference between M2M and IoT

- **Applications**
- M2M data is collected in point solutions and can be accessed by on-premises applications such as diagnosis applications, service management applications, and on-premises enterprise applications
- IoT data is collected in the cloud and can be accessed by cloud applications such as analytics applications, enterprise applications, remote diagnosis and management applications, etc

Communication - M2M vs IoT



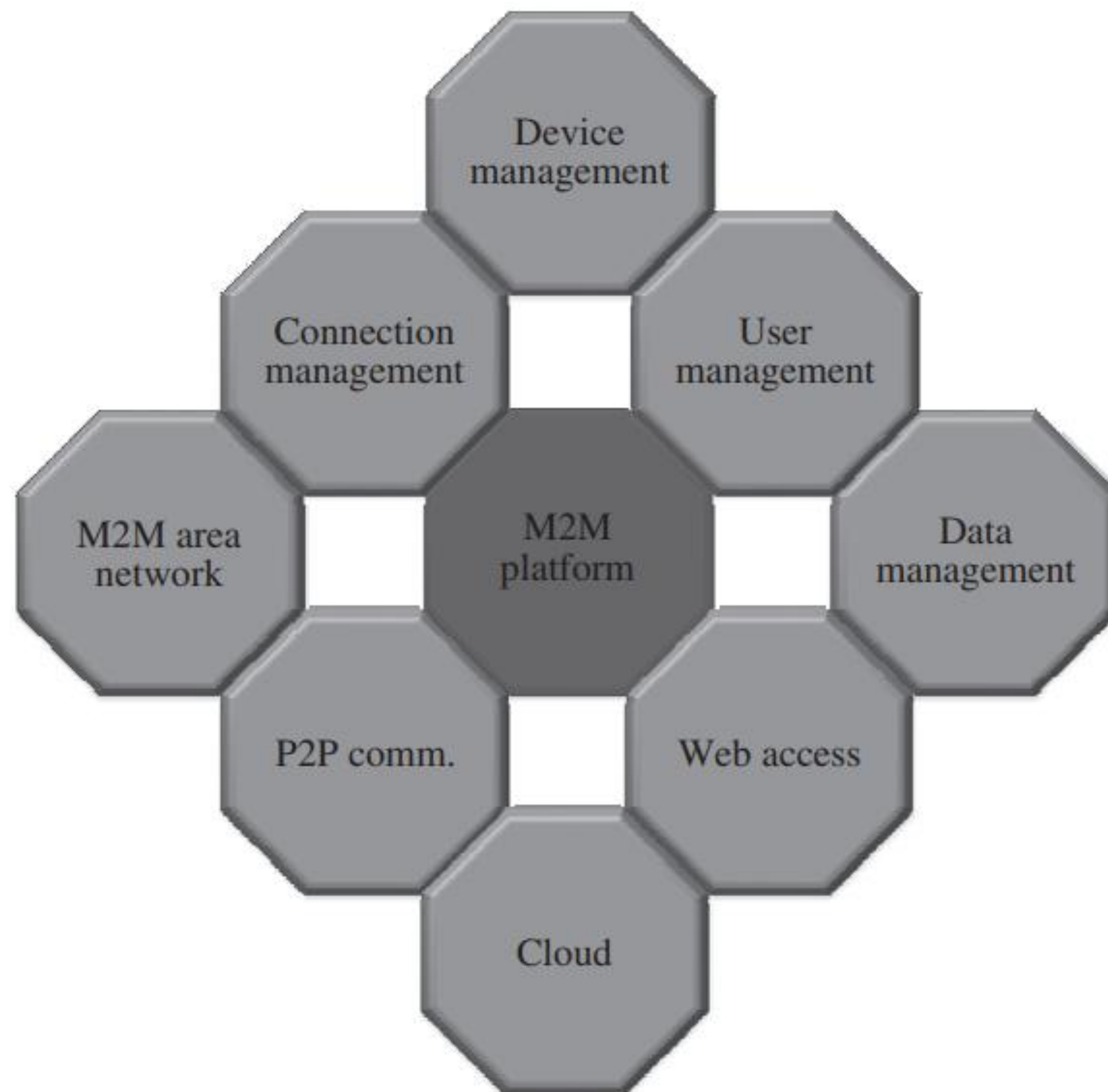
M2M Examples

- ATM machines signaling banks about the need to refill them with cash
- Power line monitoring systems in a house alerting a generator set of possible power failures and switching to generator-based supply
- A common example of a machine to machine is controlling electrical devices like fans and bulbs using Bluetooth from the smartphone.
- Here, the smartphone and electrical devices are the two interacting devices with each other

M2M Characteristics

- **Significant characteristics of M2M:**
- Heterogeneous markets
- Low data footprint
- Low cost, maintenance
- Packet Switched
- Event Detection
- Very large number of communicating devices working without human intervention

M2M Features



M2M Architecture

- Networking Model
 - Devices
 - Network
 - Gateway
- Service Model
 - Device Management
 - User Management
 - Data and Analytics Platform